Ordinary Differential Equations

Lecture Notes (LaTeX edition)

October 2026

Abstract

These notes are transcribed from my handwritten notebook on Ordinary Differential Equations. They cover: explicit solution methods for first order ODEs (exact equations, separating variables, integrating factors, implicit equations); existence, uniqueness and extension of solutions (Gronwall, Arzelà–Ascoli, Picard–Lindelöf, Peano, Osgood, comparison principles, dependence on data and parameters); linear systems (fundamental matrices, constant coefficients, higher order equations, Floquet theory); the power series method (Cauchy theory, Frobenius theory, Bessel functions); and Sturm–Liouville theory (separation and comparison theorems, the regular and periodic Sturm–Liouville problems, boundary value problems for nonlinear ODEs).

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Chapter 1 Introduction

1.1 What is an ODE

Def ((ODE)). F is a given function, $x \in \mathbb{R}$, $y(x)\colon I \subseteq \mathbb{R} \to V \subseteq \mathbb{R}^m$ $(m < \infty)$. An ODE is

$$F(x, y(x), y'(x), \dots, y^{(n)}(x)) = 0,$$

which has order n.

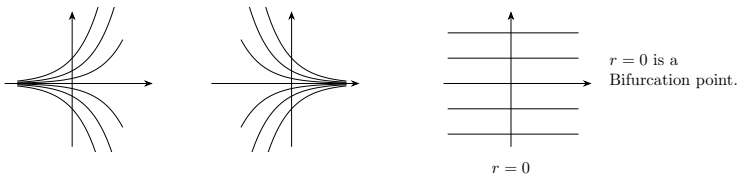
Def ((Classical solution)). $\varphi(x)\colon J \subseteq \mathbb{R} \to V$, $\varphi \in D^n(J)$, s.t.

$$F(x, \varphi(x), \varphi'(x), \dots, \varphi^{(n)}(x)) = 0, \quad \forall x \in J.$$

Rem. If F is good enough, by implicit function theorem,

$$y^{(n)} = f(x, y, y', \dots, y^{(n-1)}).$$

Notes for example 1.5: long time behavior.



$r = 0$ is a **bifurcation point**.

Def ((General solution)). For $F(x, y, y', \dots, y^{(n)}) = 0$, if $\varphi(x, C_1, \dots, C_n)$ solves it, and C_i are **independent parameters**, we say φ is a general solution.

Here, "independent parameters" means

$$\det \frac{\partial(\varphi, \varphi', \dots, \varphi^{(n-1)})}{\partial(C_1, C_2, \dots, C_n)} \neq 0.$$

Rem. If φ is a general solution, then given any initial data $(y(0), y'(0), \dots, y^{(n-1)}(0)) = (a_0, a_1, \dots, a_{n-1})$, we can always find some C_i s.t. $\varphi(x, C_1, \dots, C_n)$ solves the IVP. This is true by the inverse function theorem.

1.2 Find explicit solutions for ODE

1.2.1 Exact equation

Consider $P(x, y)\,dx + Q(x, y)\,dy = 0$.

Def ((general integral)). If $\exists\ \Phi(x, y) \in C^1$ s.t.

$$d\Phi = P(x, y)\,dx + Q(x, y)\,dy,$$

then the equation is called an **exact equation**, and Φ is called a general integral of the equation. Its solutions are given by $\Phi(x, y) = C$.

Lemma ((Exact-equation lemma; proof see the lecture notes)). Assume P, Q and $\partial P/\partial y$, $\partial Q/\partial x$ are continuous in a simply connected domain. Then the equation is exact iff

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

In this case,

$$\Phi(x,y) = \int_{x_0}^{x} P(s,y)\,ds + \int_{y_0}^{y} Q(x,\theta)\,d\theta + C.$$

Rem. Compare with: "$\int_\gamma P\,dx + Q\,dy$ is independent of path iff $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ (for a simply connected domain)."

1.2.2 Separating variables

If $P(x,y) = X(x)Y(y)$, $Q(x,y) = X_1(x)Y_1(y)$, then we rewrite

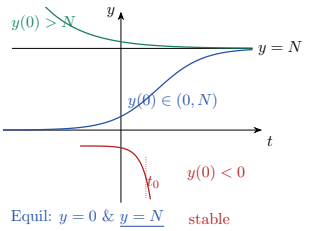
$$\frac{X(x)}{X_1(x)}\,dx + \frac{Y_1(y)}{Y(y)}\,dy = 0,$$

which is obviously exact. Notice $X_1(x) = 0$ (i.e. $x = a_1$) or $Y_1(y) = 0$ (i.e. $y = b_1$), which are also solutions.

Eg ((Logistic population model)). $\frac{dy}{dt} = ry(N-y)$

$$\Rightarrow \quad \frac{dy}{y(N-y)} = r\,dt \quad (\text{or } y \equiv 0,\ y \equiv N) \quad \Rightarrow \quad \frac{y}{N-y} = k\,e^{Nrt} \quad (k \in \mathbb{R})$$

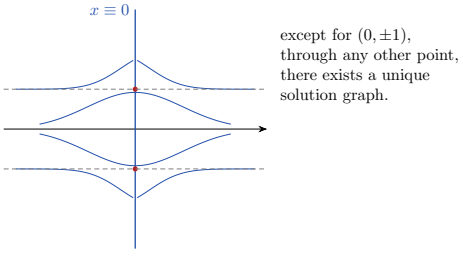
$$\Rightarrow \quad y(t) = \frac{Nke^{Nrt}}{1+ke^{Nrt}} = N - \frac{N(N-y(0))}{(N-y(0)) + y(0)\,e^{Nrt}}.$$



Equilibrium: $y = 0$ & $y = N$ (stable).

Eg. $(x^2+1)(y^2-1)\,dx + xy\,dy = 0$

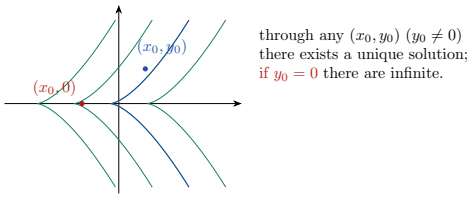
$$\Rightarrow \quad y^2 - 1 = \frac{k}{x^2 e^x} \quad (k \in \mathbb{R}), \qquad \& \quad x \equiv 0.$$



Except for $(0, \pm 1)$, through any other point there exists a unique solution graph.

Eg. $y' = \frac{3}{2}y^{1/3}$

$$\Rightarrow\ y^2 = (x+C)^3 \quad \& \quad y \equiv 0.$$



Through any (x_0, y_0) with $y_0 \neq 0$ there exists a unique solution; if $y_0 = 0$ there are infinitely many.

Eg ((Homogeneous equation of order n))**.** We call $f(x,y)$ a homogeneous function of order n if $f(tx,ty) = t^n f(x,y)$ for all t.

Consider $M(x,y)\,dx + N(x,y)\,dy = 0$, in which M, N are homogeneous of order n. Set $y = ux$. Then $M(x,y) = x^n M(1,u)$, $dy = x\,du + u\,dx$,

$$x^n\big(M(1,u) + uN(1,u)\big)\,dx + x^{n+1}N(1,u)\,du = 0.$$

Eg. $\frac{dy}{dx} = \frac{x+y}{x-y}$. Set $y = ux$, $dy = u\,dx + x\,du$:

$$\frac{u\,dx + x\,du}{dx} = \frac{x+ux}{x-ux} = \frac{1+u}{1-u} \ \Rightarrow\ \frac{(1-u)\,du}{1+u^2} = \frac{dx}{x}$$

(here, check that $x \equiv 0$ is not a solution)

$$\Rightarrow\ \arctan u - \tfrac{1}{2}\ln(1+u^2) = \ln|x| + C.$$

1.2.3 Integrating factor

Find a function $a(x,y)$ s.t. $aP(x,y)\,dx + aQ(x,y)\,dy = 0$ is exact. Then $\exists\ \Phi$ s.t. $d\Phi = aP\,dx + aQ\,dy$,

$$\frac{\partial(aP)}{\partial y} = \frac{\partial^2\Phi}{\partial x\,\partial y} = \frac{\partial(aQ)}{\partial x}.$$

Lemma. $P\,dx + Q\,dy = 0$ admits an integrating factor of the form $a(x)$ iff $G = \big(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\big)/Q$ is independent of y; in this case $a(x) = e^{\int_{x_0}^{x} G(t)\,dt}$.

Similarly, it admits an integrating factor of the form $a(y)$ iff $H = \big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\big)/P$ is independent of x; in this case $a(y) = e^{\int_{y_0}^{y} H(t)\,dt}$ is an integrating factor.

Eg. $(3x^2y + 2xy + y^3)\,dx + (x^2 + y^2)\,dy = 0$:

$$\frac{\partial P}{\partial y} = 3x^2 + 2x + 3y^2, \qquad \frac{\partial Q}{\partial x} = 2x, \qquad \frac{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}}{Q} = 3.$$

Take e^{3x} as the integrating factor.

Eg (Back to the homogeneous equation)**.**

$$x^n\big(M(1,u) + uN(1,u)\big)\,dx + x^{n+1}N(1,u)\,du = 0.$$

Notice $\dfrac{1}{x^{n+1}(M(1,u) + uN(1,u))} = \dfrac{1}{xM(x,y) + yN(x,y)}$; we guess this is an integrating factor, i.e. $\dfrac{M}{xM+yN}\,dx + \dfrac{N}{xM+yN}\,dy$ is exact.

Pf. $\partial_y\Big(\dfrac{M}{xM+yN}\Big) - \partial_x\Big(\dfrac{N}{xM+yN}\Big) = \frac{1}{(xM+yN)^2}\Big[N(xM_x + yM_y) - M(xN_x + yN_y)\Big] = 0.$ $\square$

1.2.4 First order equation

Consider $\frac{dy}{dx} + p(x)y - q(x) = 0$, i.e.

$$\big(p(x)y - q(x)\big)\,dx + dy = 0. \qquad (*)$$

Taking the integrating factor $a(x) = e^{\int_{x_0}^x p(s)\,ds}$,

$$e^{\int_{x_0}^x p(s)\,ds}\big(p(x)y - q(x)\big)\,dx + e^{\int_{x_0}^x p(s)\,ds}\,dy = 0 \;\Rightarrow\; e^{\int_{x_0}^x p(s)\,ds}\cdot y - \int_{x_0}^x e^{\int_{x_0}^t p(s)\,ds}\,q(t)\,dt = C,$$

$$\Rightarrow\; y = e^{-\int_{x_0}^x p(s)\,ds}\Big(C + \int_{x_0}^x q(t)\,e^{\int_{x_0}^t p(s)\,ds}\,dt\Big).$$

Prop. Every $y(x)$ solving $(*)$ takes the form above.

Pf. $\Big(y(x)e^{\int_{x_0}^x p(s)\,ds}\Big)' = e^{\int_{x_0}^x p(s)\,ds}(y'(x) + p(x)y(x)) = e^{\int_{x_0}^x p(s)\,ds}\,q(x)$. □

Lemma. (1) For the homogeneous equation $y' + p(x)y = 0$, the solutions form a 1-dim vector space.

(2) For the inhomogeneous equation, the general solution consists of two parts: one is the general solution of the homogeneous equation, the other is a special solution.

(3) The first order equation has a unique solution for the IVP.

Eg. $\frac{dy}{dx} + ay = f(x)$, where $f(x + 2\pi) = f(x)$, $x \in \mathbb{R}$. Find a 2π-periodic solution.

Sol. Claim: if we have a solution s.t. $y(0) = y(2\pi)$, then $y(x)$ is 2π-periodic. Indeed $y(x)$ is a solution $\Rightarrow y(x + 2\pi)$ is also a solution; then $z(x) = y(x + 2\pi) - y(x)$ solves $z' + az = 0$ (from the lemma), so $z(x) = e^{-ax}z(0)$; if $y(0) = y(2\pi)$ then $z(0) = 0 \Rightarrow z(x) \equiv 0 \Rightarrow y(x + 2\pi) = y(x)$ for all x.

So we just solve the equation and impose $y(0) = y(2\pi)$:

$$y' = -ay + f(x) \;\Rightarrow\; y = e^{-ax}\Big(C + \int_0^x f(s)\,e^{as}\,ds\Big),$$

$$y(0) = y(2\pi) \;\Rightarrow\; C = \frac{e^{-2\pi a}\int_0^{2\pi} f(x)\,e^{ax}\,dx}{1 - e^{-2\pi a}}.$$

1.2.5 First order ODE in implicit form

Consider $F(x, y', y) = 0$. $(*)$

Special case. If y can be solved: $y = f(x, y')$; let $p := \frac{dy}{dx}$. Then

$$p = \frac{dy}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial p}\cdot\frac{dp}{dx} \;\Rightarrow\; \frac{\partial f}{\partial p}(x,p)\,\frac{dp}{dx} = p - \frac{\partial f}{\partial x}(x,p).$$

If $p = p(x)$ is a solution, then $f(x, p(x))$ solves $(*)$.

Eg ((Clairaut equation)). $y = xp + f(p)$, $p := \frac{dy}{dx}$:

$$\frac{dy}{dx} = p + x\frac{dp}{dx} + f'(p)\frac{dp}{dx} \;\Rightarrow\; \big(x + f'(p)\big)\frac{dp}{dx} = 0,$$

$$\Rightarrow\; y = cx + f(c) \quad \text{or} \quad \begin{cases} x = -f'(p) \\ y = -f'(p)\,p + f(p) \end{cases} \qquad (p \text{ as a parameter}).$$

Method of parameter. $F(x, y, p) = 0$: take $x = x(u, v)$, $y = y(u, v)$, $p = p(u, v)$.

Eg. $x = (y')^3 + y'$. See $p = y'$ as a parameter:

$$x = p^3 + p, \qquad dy = p\,dx = p(3p^2 + 1)\,dp \;\Rightarrow\; \begin{cases} x = p^3 + p \\ y = \frac{3}{4}p^4 + \frac{1}{2}p^2 + C. \end{cases}$$

1.2.6 Other examples

Eg. $\frac{dy}{dx} = f(x+y)$: let $z = x + y(x)$, then $\frac{dz}{dx} = 1 + \frac{dy}{dx} = f(z)$.

Eg. $\frac{dy}{dx} = f\Big(\frac{ax+by+c}{mx+ky+l}\Big)$.

Case 1. $c = 0$, $l = 0$: the equation is homogeneous of order 0.

Case 2. $c\,l \neq 0$, $\Delta = \begin{vmatrix} a & b \\ m & k \end{vmatrix} \neq 0$: then $\exists\ \tilde{x} = x + x_0$, $\tilde{y} = y + y_0$ s.t. $\frac{d\tilde{y}}{d\tilde{x}} = f\Big(\frac{a\tilde{x}+b\tilde{y}}{m\tilde{x}+k\tilde{y}}\Big)$.

Case 3. $c\,l \neq 0$, $\Delta = 0$: then $f\Big(\frac{ax+by+c}{mx+ky+l}\Big) = f\Big(\frac{ax+by+c}{\lambda(ax+by)+l}\Big)$; taking $z = ax + by$ we can separate variables.

Eg ((Bernoulli eq.)). $\frac{dy}{dx} + p(x)y = q(x)y^n$

$$\Rightarrow\ \frac{1}{y^n}\frac{dy}{dx} + p(x)y^{1-n} = q(x); \qquad z = y^{1-n}(x)\ \Rightarrow\ \frac{dz}{dx} = (1-n)\big(q(x) - p(x)z\big),$$

a first order equation for $z(x)$.

Eg ((Riccati eq.)). $\frac{dy}{dx} = a(x)y^2 + b(x)y + c(x)$.

Lemma. (1) If $\Phi(x)$ is a solution, then we can find all solutions.

Pf. Assume $y(x) = \Phi(x) + u(x)$:

$$y' = \Phi'(x) + u'(x) = a(\Phi+u)^2 + b(\Phi+u) + c = \Phi' + 2a\Phi u + au^2 + bu,$$

$$\Rightarrow\ \frac{du}{dx} = (2a\Phi + b)u + au^2 \qquad \text{(Bernoulli eq.)}$$

(2) If φ_1, φ_2 are two given solutions, then any solution $y(x)$ satisfies

$$\frac{y-\varphi_1}{y-\varphi_2} = C\exp\Big(\int_{x_0}^{x} a(s)(\varphi_1-\varphi_2)(s)\,ds\Big).$$

(3) If $\varphi_1, \varphi_2, \varphi_3$ are solutions, then

$$\frac{y-\varphi_1}{y-\varphi_2}\Big/\frac{\varphi_3-\varphi_1}{\varphi_3-\varphi_2} = C.$$

Thm ((Bernoulli & Liouville)). $\frac{dy}{dx} = ay^2 + bx^m$ $(a, b, m \in \mathbb{R},\ a \neq 0)$ can be solved iff

$$m = 0, \quad -2, \quad \frac{-4k}{2k+1}, \quad \frac{-4k}{2k-1} \quad (k \in \mathbb{Z}).$$

Chapter 2 Existence & Uniqueness

2.1 Preparation

2.1.1 Gronwall inequality

Lemma (Version 1). $f \in C^1[a,b]$, $g \in C[a,b]$, and $f' \le gf$ for all $x \in [a,b]$. Then

$$f(x) \le f(x_0)\exp\Big(\int_{x_0}^{x} g(s)\,ds\Big), \qquad \forall\, x \ge x_0,\ x_0 \in [a,b].$$

Pf. $\big(e^{-\int_{x_0}^{x} g(s)\,ds} f\big)' = e^{-\int_{x_0}^{x} g(s)\,ds}(f' - gf) \le 0$; then $e^{-\int_{x_0}^{x} g(s)\,ds} f(x) \le f(x_0)$. □

Lemma (Version 2). $f \in C^1[a,b]$, $g, h \in C[a,b]$, $f' \le h + gf$. Then

$$f(x) \le e^{\int_{x_0}^{x} g(s)\,ds}\Big(f(x_0) + \int_{x_0}^{x} e^{-\int_{x_0}^{t} g(s)\,ds} h(t)\,dt\Big), \qquad \forall\, x \ge x_0.$$

Pf. $\big(e^{-\int_{x_0}^{x} g(s)\,ds} f(x)\big)' \le e^{-\int_{x_0}^{x} g(s)\,ds} h(x)$; then apply Version 1. □

Lemma (Version 3). $f, g \in C[a,b]$, $g \ge 0$, $f(x) \le C + \int_a^x g(s) f(s)\,ds$. Then $f(x) \le C\exp\big(\int_a^x g(s)\,ds\big)$.

Pf. $\Phi(x) := C + \int_a^x g(s) f(s)\,ds$, $\Phi'(x) = g(x) f(x) \le g(x)\Phi(x)$; $\Rightarrow f(x) \le \Phi(x) \overset{\text{Ver 1}}{\le} C\exp\big(\int_a^x g(s)\,ds\big)$. □

Lemma (Version 4). $f, g, h \in C[a,b]$, $g \ge 0$, $f(x) \le h(x) + \int_a^x g(s) f(s)\,ds$. Then

$$f(x) \le h(x) + e^{\int_a^x g(s)\,ds}\Big[\int_{x_0}^{x} e^{-\int_{x_0}^{t} g(s)\,ds} g(t)\,h(t)\,dt\Big].$$

Pf. $\Phi(x) := \int_a^x g(s) f(s)\,ds$, $\Phi' = gf \le g(h + \Phi) = gh + g\Phi$; $\Rightarrow f(x) \le h(x) + \Phi(x) \overset{\text{Ver 2}}{\le} \dots$ □

2.1.2 Ascoli–Arzelà lemma

Def. $f_n \in C[a,b]$ are called **uniformly bounded** if $|f_n(x)| < K$ for all x, n.

Def. $f_n \in C[a,b]$ are called **equicontinuous** if $\forall \varepsilon > 0\ \exists\, \delta$ s.t. $|x - y| < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$ for all n.

Lemma ((Arzelà–Ascoli)). If $f_n \in C[0,1]$ are uniformly bounded & equicontinuous, then $\exists$ a subsequence $f_{n_j} \Rightarrow f \in C[a,b]$.

Pf. Let $\{x_n\} := \mathbb{Q} \cap [a,b]$. For x_1, $\{f_n(x_1)\}$ is bounded, so there is a subsequence $f_{1,n}(x_1)$ converging. For x_2: $\{f_{1,n}(x_2)\}$ is bounded, take a further subsequence $\{f_{2,n}(x_2)\} \subseteq \{f_{1,n}(x_2)\}$ converging. Repeating the process gives $\{f_{k,n}\} \subseteq \{f_{l,n}\}$ for $l \le k-1$. Take the diagonal $\{f_{n,n}\}$: $f_{n,n}(x_j)$ converges for every rational x_j.

Since $\{f_n\}$ is equicontinuous, $\forall \varepsilon > 0\ \exists \delta > 0$: $|x-y| < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon/3$. Cover $[a,b] \subseteq \bigcup_{k=1}^{K} B(x_{j_k}, \delta)$; $\forall \varepsilon > 0\ \exists N = N(\varepsilon, j_k)$ s.t. $|f_{n,n}(x_{j_k}) - f_{m,m}(x_{j_k})| < \varepsilon$. Take $N_0 = \max_k N(\varepsilon, j_k)$; then for $n, m > N_0$:

$$|f_{n,n}(x) - f_{m,m}(x)| = |f_{n,n}(x) - f_{n,n}(x_j)| + |f_{n,n}(x_j) - f_{m,m}(x_j)| + |f_{m,m}(x_j) - f_{m,m}(x)| < \varepsilon.$$

Hence $f_{n,n}$ converges uniformly. □

Rem. If M is a compact set of $\mathbb{R}^n$, the A–A thm. applies to $\{f_n\} \subseteq C(M, \mathbb{R})$. Also, the A–A thm applies to $C(X, Y)$ where X, Y are complete metric spaces, X compact and $\{f_n\}$ totally bounded.

2.2 Picard–Lindelöf thm.

Consider the IVP

$$\begin{cases} \dfrac{dy}{dx} = f(x,y) \\ y(x_0) = y_0 \end{cases} \qquad D = \big\{(x,y) \in \mathbb{R}^2 :\ |x-x_0| < a,\ |y-y_0| < b\big\}.$$

Def ((Lipschitz)). f is called y-Lipschitz on D if $|f(x,y_1) - f(x,y_2)| < L\,|y_1 - y_2|$ for all $(x,y_1),(x,y_2) \in D$. f is called locally Lipschitz on Ω if every $(x_0,y_0) \in \Omega$ has a neighborhood D s.t. f is Lipschitz on D.

Thm ((Picard–Lindelöf)). If $f \in C(D)$ and f is y-Lipschitz on D, then the Cauchy problem has a unique solution $y(x)$ on $[x_0 - h,\ x_0 + h]$, where $h := \min\{a,\ b/M\}$.

Pf. Step 1. Transform to the integral equation

$$y(x) = y_0 + \int_{x_0}^{x} f(s, y(s))\, ds. \qquad (*)$$

We claim ODE $\iff$ $(*)$: if $y \in C^1$ solves the ODE then y satisfies $(*)$; if $y \in C$ satisfies $(*)$ then $y \in C^1$ & solves the ODE. Denote $\Phi(y) := y_0 + \int_{x_0}^{x} f(s,y(s))\,ds$; we search for a fixed point $y = \Phi(y)$. (Formally we can use the Banach thm.)

Step 2. Construct the Picard sequence:

$$y_0(x) \equiv y_0, \quad y_1(x) := \Phi(y_0) = y_0 + \int_{x_0}^{x} f(s, y_0(s))\, ds, \quad \ldots, \quad y_n := \Phi(y_{n-1}).$$

Check: $|y_1 - y_0| = \big|\int_{x_0}^{x} f(s,y_0(s))\,ds\big| \le Mh \le b$, and $|y_n - y_0| \le b \Rightarrow |y_{n+1} - y_0| \le b$.

Step 3. Prove y_n converges uniformly on $[x_0 - h,\ x_0 + h]$:

$$|y_1(x) - y_0(x)| \le M|x - x_0|,$$

$$|y_2(x) - y_1(x)| = \Big|\int_{x_0}^{x} \big(f(s,y_1(s)) - f(s,y_0(s))\big)\Big| \le \int_{x_0}^{x} L\,|y_1(s) - y_0(s)|\,ds \le L\,M\,\frac{(x-x_0)^2}{2}.$$

Similarly,

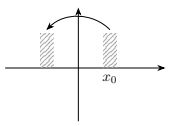
$$|y_n(x) - y_{n-1}(x)| \le M\,\frac{L^{n-1}}{n!}\,|x - x_0|^n.$$

Notice $|y_k(x) - y_{k-1}(x)| \le \frac{ML^{k-1}}{k!}h^k$, whose RHS converges; so $y_n(x) := \sum_{k=1}^{n}(y_k - y_{k-1}) + y_0$ converges uniformly. Taking the limit gives $y := \lim y_n$ with $y = \Phi(y)$; hence $y(x)$ solves the Cauchy problem.

Step 4. Uniqueness. Suppose φ_1, φ_2 solve the IVP:

$$|\varphi_1(x) - \varphi_2(x)| \le \Big|\int_{x_0}^{x} L\,|\varphi_1(s) - \varphi_2(s)|\,ds\Big|.$$

Using Gronwall: $|\varphi_1(x) - \varphi_2(x)| \le C\,e^{L(x-x_0)}$ with $C = 0$, so $\varphi_1(x) = \varphi_2(x)$ for $x \ge x_0$. For $x \le x_0$ consider $\tilde{\varphi}_i(x) = \varphi_i(-x)$, which transforms the uniqueness to the right. □



Cor ((Convergence rate)). Under the same conditions,

$$|y_n(x) - y(x)| \le \frac{ML^n}{(n+1)!}\,|x - x_0|^{n+1}, \qquad \forall\, x \in [x_0 - h,\ x_0 + h].$$

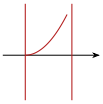
Eg. $\frac{dy}{dx} = x^2 + y^2$: for any (x_0, y_0) there is a unique solution curve through it (because $f(x,y) = x^2 + y^2$ is Lipschitz on every bounded domain).

Eg. $\frac{dy}{dx} = y^{1/3}$: for (x_0, y_0) there is a unique solution curve through it if $y_0 \neq 0$ (because $f(x,y) = y^{1/3}$ is Lipschitz on domains where $y \neq 0$).

Rem. Picard–Lindelöf holds for $y(x) : I \subseteq \mathbb{R} \to X$ with X a complete metric space.

Eg ((Miller, 1923)). $y' = f(x,y)$, $y(0) = 0$, where

$$f(x,y) = \begin{cases} 0 & , \ x = 0, \ y \in \mathbb{R} \\ 2x & , \ 0 \le x \le 1, \ y \le 0 \\ 2x - \dfrac{4y}{x} & , \ 0 < x \le 1, \ 0 < y < x^2 \\ -2x & , \ 0 < x \le 1, \ y \ge x^2 \end{cases}$$



$f(x,y)$ is continuous but not Lipschitz (notice $\frac{4|y_1 - y_2|}{|x|}$ near $x = 0$).

Picard sequence: $y_0(x) \equiv 0$, $y_1(x) = x^2$, $y_2(x) = -x^2$, $y_3(x) = x^2$, ...

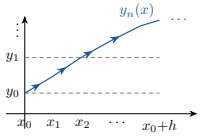
But $f(x,y)$ is decreasing in y.

2.3 Peano's thm & Osgood uniqueness

Thm ((Peano)). For $\begin{cases} y' = f(x,y) \\ y(x_0) = y_0 \end{cases}$ with $D := \{(x,y) : \ |x - x_0| \le a, \ |y - y_0| \le b\}$ and $f \in C(D)$, the Cauchy problem has **at least one solution** on $x \in [x_0 - h, \ x_0 + h]$, $h := \min\{a, \ b/M\}$, $M := \max_D f(x,y)$.

Pf. For simplicity, construct the solution on $[x_0, x_0 + h]$ only. For each n, cut $[x_0, x_0 + h]$ into n segments, $x_k := x_0 + \frac{k}{n}h$. Construct, for $x \in [x_k, x_{k+1}]$,

$$y_n(x) = y_0 + \sum_{j=0}^{k-1} f(x_j, y_j)(x_{j+1} - x_j) + f(x_k, y_k)(x - x_k), \qquad y_j := y_n(x_j).$$

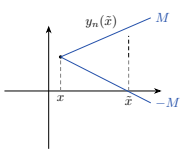


Properties of y_n:

(1) $y_n \in C[x_0, x_0 + h]$ for all n.

(2) $|y_n(x) - y_0| \le Mh$ for all n (uniformly bounded).

(3) $|y_n(x) - y_n(\tilde{x})| \le M$ for all n and $x, \tilde{x} \in [x_0, x_0 + h]$ (taking x as a starting point, $y_n(\tilde{x})$ varies by at most M).



(4) $y_n(x)$ approximates the solution:

$$\Big|\, y_n(x) - \Big[\, y_0 + \int_{x_0}^{x} f(s, y_n(s))\, ds \,\Big]\Big| \le \sum_{j=0}^{k-1} \int_{x_j}^{x_{j+1}} |f(x_j, y_j) - f(s, y_n(s))|\, ds + \int_{x_k}^{x} |f(x_k, y_k) - f(s, y_n(s))|\, ds < \varepsilon$$

(taking $n \to \infty$, using $f \in C(D) \Rightarrow$ uniformly continuous). By the A–A lemma we find $y_{n_k}(x) \rightrightarrows y(x)$ on $[x_0, x_0 + h]$; easily $\big|y(x) - [y_0 + \int_{x_0}^{x} f(s, y(s))\, ds]\big| < \varepsilon(1 + 2h)$. Hence $y(x)$ solves the equation. □

Def. $f(x, y) \in C(\Omega)$ & $|f(x, y_1) - f(x, y_2)| < F(|y_1 - y_2|)$, where $F \in C(0, +\infty)$ & $\int_0^{\varepsilon} \frac{dr}{F(r)} = +\infty$ for all $\varepsilon > 0$. Then $f(x, y)$ satisfies the **Osgood condition** on Ω w.r.t. y.

Thm ((Osgood))**.** For the equation in Peano's thm., if $f(x, y)$ satisfies the Osgood condition, then the solution is unique.

Pf. Suppose $\varphi_1(x), \varphi_2(x)$ are two different solutions with $\varphi_1(x_0) = \varphi_2(x_0)$; $\exists\, x_1$: $\varphi_1(x_1) \neq \varphi_2(x_1)$. Assume $x_1 > x_0$ & $\varphi_1(x_1) > \varphi_2(x_1)$ and set

$$r(x) := \varphi_1(x) - \varphi_2(x), \qquad x^* := \sup\{x \in [x_0, x_1] : r(x) = 0\},$$

so $r(x^*) = 0$ and $r(x) > 0$ for $x \in (x^*, x_1]$. Then

$$r'(x) = f(x, \varphi_1(x)) - f(x, \varphi_2(x)) \le F(|\varphi_1(x) - \varphi_2(x)|) = F(r(x)), \qquad x \in (x^*, x_1],$$

$$\Rightarrow\ \frac{r'(x)}{F(r(x))} \le 1\ \Rightarrow\ \int_{x^*}^{x_1} \frac{r'(x)}{F(r(x))}\, dx \le x_1 - x^*\ \Rightarrow\ +\infty \le x_1 - x^*,$$

a contradiction. □

Prop. For the equation in Peano's thm., if $f(x, y)$ is decreasing w.r.t. y, then the solution is unique.

Pf. With $\varphi_1, \varphi_2, r(x), x^*$ as above: $r'(x) = f(x, \varphi_1) - f(x, \varphi_2) \le 0$ on $[x^*, x_1]$, so $r(x) \downarrow$; but $r(x^*) = 0$ and $r(x_1) > 0$. □

Ex. $y' = \begin{cases} 0 & ,\ y = 0 \\ y \ln |y| & ,\ y \neq 0 \end{cases}$: the solution exists & is unique.

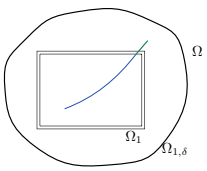
Pf. For $(x_0, y_0) \in \mathbb{R}^2$: if $y_0 \neq 0$, use Picard–Lindelöf; if $y_0 = 0$, existence holds by Peano's thm. Notice $y \equiv 0$ is a solution; if $\varphi(x)$ were another solution, then on $(x^*, x_1]$

$$\int_{x^*}^{x_1} \frac{\varphi'(x)}{\varphi(x) \ln |\varphi(x)|}\, dx = \ln\big|\ln |\varphi(x)|\big|\Big|_{\varphi(x^*)}^{\varphi(x_1)} = +\infty,$$

so the assumption is wrong. □

2.4 Extension of solutions

Thm ((Extension of solutions))**.** For $\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$ with $f \in C(\Omega)$, Ω a domain in $\mathbb{R} \times \mathbb{R}^n$: any solution curve Γ must extend to the boundary of Ω, i.e. for every compact $\Omega_1 \ni (x_0, y_0)$, Γ extends into $\Omega \setminus \Omega_1$.



Def. y_1 & y_2 are two solutions of $\begin{cases} y' = f(x,y) \\ y_0 = y(x_0) \end{cases}$: if $\text{domain}(y_2) \supseteq \text{domain}(y_1)$ and $y_2 = y_1$ on the common domain, we say y_2 is an **extension** of y_1.

Def. $y(x)$ is called a **maximal solution** if it is defined on an open interval and cannot be extended. Denote $(T_-, T_+) := \text{domain}(y)$.

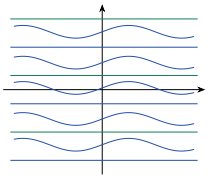
Thm. Suppose Ω is a domain and $f \in C(\Omega)$. For every (x_0, y_0) there **exists a maximal solution** of the Cauchy problem.

Pf. We prove it for the easier case (assuming local uniqueness). Let $S = \{$ all solutions on an open interval $\}$. For $\varphi \in S$ let I_φ be its open interval; denote $I = \bigcup_{\varphi \in S} I_\varphi$, clearly open. We show I is an interval: $a := \inf\{x \in I\}$, $b := \sup\{x \in I\}$; for $x^* \in (a,b)$, choose $x_1 \in I$ with $x^* < x_1 < b$ close to x^*; take $\varphi \in S$ with $x_1 \in I_\varphi$; since I_φ is an open interval, $x^* \in I_\varphi \Rightarrow x^* \in I$. Thus $I = (a,b)$. Define $y_m = \varphi$ on each I_φ (unique by assumption; otherwise use Zorn's lemma). It is easy to check that y_m is a maximal solution. □

Cor. If $I_{\max} = (T_-, T_+)$ with $T_+ < +\infty$, then for every compact CC with $[x_0, T_+] \times C \subseteq \Omega$, the solution must leave $[x_0, T_+] \times C$. (y bounded $\Rightarrow$ x extended.)

Ex. $y' = (x^2 + y^2 + 1)\sin \pi y$: any solution exists on $\mathbb{R}$ and is monotonic.

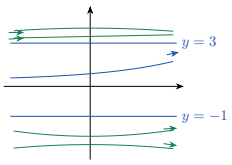
Pf. $f(x,y) \in C^1(\mathbb{R}^2)$, hence locally Lipschitz. Notice $y = n \in \mathbb{Z} \Rightarrow y' = 0$, so $y \equiv n$ are solutions.



For $y_0 = n$: the unique solution is $y \equiv n$, global. For $n < y_0 < n+1$: the unique solution cannot cross $y = n$ & $y = n+1$, hence is global. The sign of y' depends only on y, so y is monotonic. □

Ex. $y' = (y^2 - 2y - 3)\, e^{(x+y)^2}$: any solution with $I_{\max} = (T_-, T_+)$ has $T_- = -\infty$ or $T_+ = +\infty$.

Pf. $f(x,y) \in C^1(\mathbb{R})$, hence locally Lipschitz. Also $y \equiv 3$ & $y \equiv -1$ are global solutions.



For $y_0 > 3$: the solution cannot cross $y = 3$ and is increasing, so $T_- = -\infty$. For $y_0 < -1$: the solution cannot cross $y = -1$ and is increasing, so $T_+ = +\infty$. □

Ex ((Liouville)). $y' = x^2 + y^2$: any solution must have a bounded lifespan.

Pf. WLOG $x_0 \geq 0$; look at the solution φ on (T_-, T_+) and show $T_+ < +\infty$ (same for T_-). Fix $x_1 \in (x_0, T_+)$: $\varphi'(x) = x^2 + \varphi^2(x) \geq x_1^2 + \varphi^2(x)$ for $x > x_1$, so

$$\frac{\varphi'(x)}{x_1^2 + \varphi^2(x)} \geq 1 \;\Rightarrow\; \int_{x_1}^{z} \frac{\varphi'(x)}{x_1^2 + \varphi^2(x)}\, dx \geq z - x_1 \;\Rightarrow\; \frac{\pi}{x_1} \geq z - x_1 \;\Rightarrow\; z \leq x_1 + \frac{\pi}{x_1}$$

(in fact we use the comparison lemma). Letting $z \to T_+$ gives $T_+ < +\infty$. □

Ex. $\begin{cases} y' = f(x,y) \\ y_0 = y(x_0) \end{cases}$ with $f \in C(\mathbb{R}^2)$ satisfying

$$|f(x,y)| \leq M(T) + L(T)\,|y|, \qquad \forall\, T,\ \forall\, (x,y) \in [-T,T] \times \mathbb{R},$$

where $M, L > 0$ are continuous functions. Then any solution is global.

Pf. For $[x_0, T]$ with T large enough:

$$|y(x)| \le |y_0| + \int_{x_0}^{x} \big(M(T) + L(T)\,|y(s)|\big)\,ds \le \big(|y_0| + M(T)(T - x_0)\big)\,e^{L(T)(x-x_0)} \quad \text{(Gronwall)}.$$

Hence $|y(x)|$ is bounded on $[x_0, T]$ and the solution extends to $+\infty$; similarly for the negative direction. □

2.5 Comparison lemma

Lemma ((1st comparison principle)). Ω is a domain, $f, F \in C(\Omega)$ & $F > f$ on Ω (strictly). If $\begin{cases} y' = f(x,y) \\ y(x_0) = y_0 \end{cases}$ has solution φ and $\begin{cases} y' = F(x,y) \\ y(x_0) = y_0 \end{cases}$ has solution ψ on $(a,b) \ni x_0$, then $\varphi(x) > \psi(x)$ for $x \in (a, x_0)$ and $\varphi(x) < \psi(x)$ for $x \in (x_0, b)$.

Pf. $h(x) := \psi(x) - \varphi(x)$, $h(x_0) = 0$, $h'(x) > 0$ on (a,b); then $\exists\,\delta$ s.t. $h(x) > 0$ on $(x_0, x_0 + \delta)$, $h(x) < 0$ on $(x_0 - \delta, x_0)$ (using here). Suppose $h(x) > 0$ fails on (x_0, b); then $\exists\, x_1$ s.t. $h(x_1) \le 0$. Take

$$x^* = \inf\{x \in (x_0, b) : h(x) \le 0\};$$

then $h(x^*) = 0$ & $h'(x^*) < 0$, a contradiction. Similarly for the other side. □

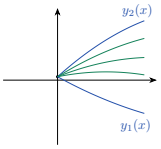
Ex. $y' = f(x,y)$ with $f \in C(\Omega)$ & $|f| \le A(x)|y| + B(x)$ on $\Omega = (a,b) \times \mathbb{R}$, where $A(x), B(x) > 0$ are continuous on (a,b). Then any solution has lifespan $I_{\max} = (a,b)$. (We don't know whether the solution is unique, so we construct objects to compare.)

Pf. Let $F := A(x)|y| + B(x) + 1$ and $G := -F$. The equation with F has a unique solution (F locally Lipschitz), which is strictly increasing, thus has at most one zero on $I_{\max}$. Case 1: no zero; WLOG $y(x) > 0$, then $y' = A(x)y + B(x)$, $y(x_0) = y_0$ is a first order ODE with a solution on (a,b). Case 2: one zero x_1; derive two equations

$$\begin{cases} y' = A(x)y + B(x) \\ y(x_1) = 0 \end{cases} \text{ on } x > x_1, \qquad \begin{cases} y' = -A(x)y + B(x) \\ y(x_1) = 0 \end{cases} \text{ on } x < x_1,$$

each with a solution on (a,b). Same for G. By the 1st comparison lemma the solution of the original equation is bounded on every compact $C \subseteq (a,b)$, thus global. □

Def. Consider $y' = f(x,y)$ with $f \in C(\Omega)$; a solution exists by Peano's thm. If y_1, y_2 are two solutions on an interval I satisfying $\forall$ solutions $y(x)$ on I: $y_1(x) \le y(x) \le y_2(x)$, we call $y_2(x)$ a **sup-solution** and $y_1(x)$ a **sub-solution**.



Thm. $f(x,y) \in C(D)$; then $\exists\,\tau > 0$ s.t. a sup-solution and a sub-solution exist on $\{x : |x - x_0| < \tau\}$.

Pf. (Create f_n to use the comparison principle.) Set $f_n(x,y) = f(x,y) + \varepsilon_n$, $\varepsilon_n \downarrow 0+$. By Peano's thm the problem $y' = f_n(x,y)$, $y(x_0) = y_0$ has a solution $\varphi_n(x)$ on $[x_0 - h_n, x_0 + h_n]$ with

$$h_n := \min\Big\{a,\ \frac{b}{M_n}\Big\}, \qquad M_n := \sup_D |f_n(x,y)|.$$

$|\varphi_n(x) - y_0| \le b$ for all n (uniformly bounded), and

$$|\varphi_n(x) - \varphi_n(x')| = \Big|\int_x^{x'} f_n(s, y_n(s))\,ds\Big| \le M_n|x - x'| \le (M+1)|x - x'|$$

(equicontinuous). Take $\tau < h$; then $\{|x-x_0| < \tau\} \subseteq \{|x-x_0| \le h_n\}$ for n large. Apply A–A on $[x_0-\tau, x_0+\tau]$: $\exists\ \varphi_{n_k} \Rightarrow \varphi$, where φ solves the original IVP. Since $f_n = f + \varepsilon_n \to f$, we get $\varphi_n(x) > \varphi(x)$ for $x > x_0$ and $\varphi_n(x) < \varphi(x)$ for $x < x_0$. Taking the h_k limit: φ is a sup-solution on $(x_0, x_0+\tau)$ and a sub-solution on $(x_0-\tau, x_0)$. Redo with $\varepsilon_n \uparrow 0-$ for the other side. □

Thm ((2nd comparison))**.** Consider

$$\begin{cases} y' = f(x,y) \\ y(x_0) = y_0 \end{cases} \quad (\times) \qquad \begin{cases} y' = F(x,y) \\ y(x_0) = y_0 \end{cases} \quad (\times\times) \qquad f(x,y) \le F(x,y) \text{ (not strictly).}$$

Let $\varphi(x)$ be a sub-solution on $[x_0, b)$ & sup-solution on $(a, x_0]$ for $(\times)$; let $\psi(x)$ be a sup-solution on $[x_0, b)$ & sub-solution on $(a, x_0]$ for $(\times\times)$. Then $\varphi(x) \le \psi(x)$ for $x \ge x_0$ and $\varphi(x) \ge \psi(x)$ for $x \le x_0$.

Pf. Solve $y' = F(x,y) + \varepsilon_n$, $y(x_0) = y_0$ ($\varepsilon_n \downarrow 0$) to get $\psi_n(x)$; solve $y' = f(x,y) - \varepsilon_n$ to get $\varphi_n(x)$ (we have to construct a strict inequality). From the existence of sup & sub solutions, $\psi_{n,k} \Rightarrow \psi$ and $\varphi_{n,k} \Rightarrow \varphi$. By the 1st comparison lemma we obtain the claim. □

Ex (starred)**.** $y' = x^2 + (y+1)^2$, $y(0) = 0$: the lifespan (T_-, T_+) satisfies $T_+ \in (\pi/4,\ 1)$.

Pf. The solution is unique and has bounded lifespan. **Step 1.** $T_+ \in [\pi/4, 1]$: consider $|x| \le 1$ and compare with $y' = (y+1)^2$ and $y' = 1 + (y+1)^2$, $y(0) = 0$:

$$\frac{-1}{x-1} - 1 \le y(x) \le \tan\Big(x + \frac{\pi}{4}\Big) - 1 \qquad \text{on } [0,1],$$

where $\frac{-1}{x-1} - 1$ exists on $[0,1)$ and $\tan(x+\frac{\pi}{4}) - 1$ exists on $[0, \frac{\pi}{4})$. Hence $T_+ \in [\pi/4, 1]$. **Step 2.** $T_+ \in (\pi/4, 1)$: notice $y' > 0$, so $x = \varphi(y)$ with $\varphi'(y) = \frac{1}{x^2+(y+1)^2}$ and $\lim_{y\to+\infty} \varphi(y) = T_+$; thus

$$T_+ = \int_0^{+\infty} \frac{ds}{x^2 + (s+1)^2}, \qquad \int_0^{+\infty} \frac{ds}{1+(s+1)^2} < T_+ < \int_0^{+\infty} \frac{ds}{(s+1)^2},$$

using $0 \le x < T_+ \le 1$. Hence $\frac{\pi}{4} < T_+ < 1$. □

2.6 Dependence on data & parameter

Q. Does a small perturbation of the data / parameter yield a small / big difference of the solution?

We can always combine the two problems. Original problem: $\begin{cases} y' = f(x,y,\lambda) \\ y(x_0) = y_0 \end{cases}$.

Reduction #1. $u := y - y_0$, $t := x - x_0$:

$$\begin{cases} u'(t) = g(u,t,\Lambda) \\ u(0) = 0 \end{cases}, \qquad \Lambda := (x_0, y_0, \lambda).$$

Reduction #2. Consider $U(x) = (y(x) \ \ z(x))^T$ with $\begin{cases} y' = f(x,y,\lambda) \\ y(x_0) = y_0 \end{cases}$ and $\begin{cases} z' = 0 \\ z(x_0) = \lambda \end{cases}$;

then $U'(x) = F(x,U) = \begin{pmatrix} f(x,y,z) \\ 0 \end{pmatrix}$, $U(x_0) = \begin{pmatrix} y_0 \\ \lambda \end{pmatrix}$.

So we should discuss equations with $y \in \mathbb{R}^n$: $y : \mathbb{R} \to \mathbb{R}^n$, $f : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$, $\begin{cases} y' = f(x,y) \\ y(x_0) = y_0 \end{cases}$,

satisfying (1) Picard–Lindelöf, (2) Peano, (3) Extension — all proved. And:

(4) $y' = A(x)y + b(x)$ with $A(x), b(x) \in C(a,b)$: the solution is unique and extends to (a,b).

Pf. It suffices to show $y(x)$ is bounded on $(c,d) \subseteq (a,b)$:

$$\frac{d}{dx}\|y(x)\|^2 = y(x)^T\big(A(x)y + b(x)\big) + \big(y^T A^T + b^T\big)y(x) \le 2\,\|y(x)\|\,\big(\|A(x)\|\,\|y(x)\| + \|b(x)\|\big),$$

$$\Rightarrow\ \frac{d}{dx}\|y(x)\| \le \|A(x)\|\,\|y(x)\| + \|b(x)\| \le M\|y(x)\| + N.$$

By Gronwall, $\|y(x)\|$ is bounded. □

2.6.1 Continuous dependence

Thm ((Continuous dependence)). Ω a bounded region of $\mathbb{R}\times\mathbb{R}^n$, $\begin{cases} y' = f(x,y) \\ y(x_0) = y_0 \end{cases}$ with $f \in C(\Omega)$; assume the solution is unique on a closed interval I_0 for every (x_0, y_0). Then the solution $\varphi(x\,;\,x_0,y_0)$ is continuous w.r.t. (x_0,y_0): $\forall\varepsilon>0\ \exists\delta>0$ s.t. $\|x_0-\tilde{x}_0\|+\|y_0-\tilde{y}_0\|<\delta$ implies

$$\|\varphi(x\,;\,x_0,y_0)-\varphi(x\,;\,\tilde{x}_0,\tilde{y}_0)\|<\varepsilon, \qquad \forall x\in I_0.$$

Pf. Suppose not: $\exists\varepsilon_0>0$ and, for all n, points with $\|(x_0^{(n)},y_0^{(n)})-(x_0,y_0)\|<\frac{1}{n}$ and $\|\varphi(x_n;x_0^{(n)},y_0^{(n)})-\varphi(x_n;x_0,y_0)\|\ge\varepsilon_0$. Since I_0 is compact, $\exists\{x_{n_k}\}\subseteq\{x_n\}$: $x_{n_k}\Rightarrow x^*\in I_0$.

Check: $\varphi(x\,;\,x_0^{(n)},y_0^{(n)})$ is uniformly bounded & equicontinuous: (1) $\varphi=y_0^{(n)}+\int_{x_0^{(n)}}^{x} f(s,\varphi(s))\,ds$ is uniformly bounded; (2) $|\varphi(x)-\varphi(\tilde{x})|=\left|\int_x^{\tilde{x}} f(s,\varphi(s))\,ds\right|\le M|x-\tilde{x}|$.

By the A–A lemma, $\varphi(x\,;\,x_0^{(n_k)},y_0^{(n_k)})\Rightarrow\widetilde{\varphi}(x)$. The IVP has a unique solution, so $\widetilde{\varphi}(x)=\varphi(x\,;\,x_0,y_0)$. Taking the limit in x:

$$0=\left\|\varphi(x^*\,;\,x_0,y_0)-\varphi(x^*\,;\,x_0,y_0)\right\|\ge\varepsilon_0,$$

a contradiction. □

Lemma ((Quantitative bound)). For $\begin{cases} y' = f(x,y) \\ y(x_0) = y_0 \end{cases}$ and $\begin{cases} y' = f(x,y) \\ y(x_0) = \tilde{y}_0 \end{cases}$ with corresponding φ,ψ on I:

$$|\varphi(x)-\psi(x)|\le|y_0-\tilde{y}_0|\,e^{L|x-x_0|},$$

where L is a Lipschitz constant of f.

Pf. WLOG $x_0=0$: $|\varphi-\psi|=\left|y_0-\tilde{y}_0+\int_{x_0}^{x} f(s,\varphi(s))-f(s,\psi(s))\,ds\right|$; apply Gronwall. □

2.6.2 C^1 dependence on data & parameters

Thm ($(C^1$ dependence)). $\begin{cases} y' = f(x,y,\lambda) \\ y(0) = 0 \end{cases}$ with $\Omega=\{\|x\|\le a,\ \|y\|\le b,\ \|\lambda\|\le c\}$, $f\in C(\Omega)$, $f_y, f_\lambda\in C(\Omega)$. Then the solution $y=\varphi(x,\lambda)\in C^1(D)$, where

$$D:=\{|x|\le h,\ |\lambda|<c\}, \qquad h:=\min\Big\{a,\ \frac{b}{M_f}\Big\}.$$

Pf. $\varphi(x,\lambda)$ is C^1 in x; we only need λ. $\varphi(x,\lambda)=\int_0^x f(s,\varphi(s,\lambda),\lambda)\,ds$. If φ were differentiable in λ, then $u:=\frac{\partial\varphi}{\partial\lambda}$ would satisfy the first order ODE

$$\begin{cases} \dfrac{du}{dx}=\dfrac{\partial f}{\partial y}(x,\varphi(x,\lambda),\lambda)\,u+\dfrac{\partial f}{\partial\lambda}(x,\varphi(x,\lambda),\lambda) \\ u(0,\lambda)=0 \end{cases} \qquad (\times) \tag{1}$$

So we only have to prove: ① $(\times)$ has a unique solution on $|x|\le h$, $|\lambda|\le C$ (first order ODE theory); ② the solution of $(\times)$ is the derivative, i.e.

$$|\varphi(x,\lambda)-\varphi(x,\lambda_0)-u(x,\lambda_0)(\lambda-\lambda_0)|=o(|\lambda-\lambda_0|).$$

For ②, expand:

$$
\begin{aligned}
&\varphi(x,\lambda)-\varphi(x,\lambda_0)-u(x,\lambda_0)(\lambda-\lambda_0)\\
&=\int_0^x f(s,\varphi(s,\lambda),\lambda)\,ds-\int_0^x f(s,\varphi(s,\lambda_0),\lambda_0)\,ds\\
&\quad-\int_0^x\Big[\frac{\partial f}{\partial y}(s,\varphi(s,\lambda_0),\lambda_0)\,u(s,\lambda_0)+\frac{\partial f}{\partial\lambda}(s,\varphi(s,\lambda_0),\lambda_0)\Big](\lambda-\lambda_0)\,ds\\
&=\int_0^x\int_0^1\frac{\partial f}{\partial\lambda}\big(s,\varphi(s,\lambda),\,\theta\lambda+(1-\theta)\lambda_0\big)d\theta\,(\lambda-\lambda_0)\,ds+\int_0^x\int_0^1\frac{\partial f}{\partial y}\big(s,\,\theta\varphi(s,\lambda)+(1-\theta)\varphi(s,\lambda_0),\,\lambda_0\big)d\theta\,\big(\varphi(s,\lambda)-\\
&\quad-\int_0^x\frac{\partial f}{\partial y}(s,\varphi(s,\lambda_0),\lambda_0)\,u(s,\lambda_0)\,(\lambda-\lambda_0)\,ds-\int_0^x\frac{\partial f}{\partial\lambda}(s,\varphi(s,\lambda_0),\lambda_0)\,(\lambda-\lambda_0)\,ds\\
&=\int_0^x\int_0^1\Big[\frac{\partial f}{\partial\lambda}\big(s,\varphi(s,\lambda),\theta\lambda+(1-\theta)\lambda_0\big)-\frac{\partial f}{\partial\lambda}(s,\varphi(s,\lambda_0),\lambda_0)\Big]d\theta\,ds\,(\lambda-\lambda_0)\\
&\quad+\int_0^x\int_0^1\frac{\partial f}{\partial y}\big(s,\theta\varphi(s,\lambda)+(1-\theta)\varphi(s,\lambda_0),\lambda_0\big)d\theta\,\big[\varphi(s,\lambda)-\varphi(s,\lambda_0)-u(s,\lambda_0)(\lambda-\lambda_0)\big]ds\\
&\quad+\int_0^x\int_0^1\Big[\frac{\partial f}{\partial y}\big(s,\theta\varphi(s,\lambda)+(1-\theta)\varphi(s,\lambda_0),\lambda_0\big)-\frac{\partial f}{\partial y}(s,\varphi(s,\lambda_0),\lambda_0)\Big]d\theta\,u(s,\lambda_0)(\lambda-\lambda_0)\,ds.
\end{aligned}
$$

Denote $\Delta(x)=|\varphi(x,\lambda)-\varphi(x,\lambda_0)-u(x,\lambda_0)(\lambda-\lambda_0)|$. By continuity of $\frac{\partial f}{\partial y}$, $\frac{\partial f}{\partial\lambda}$ and $\varphi(x,\lambda)$:

$$
\Delta(x)\le\Big|\int_0^x\int_0^1\varepsilon_1\,d\theta\,ds\,(\lambda-\lambda_0)\Big|+\Big|\int_0^x\int_0^1K\,d\theta\,\Delta(s)\,ds\Big|+\Big|\int_0^x\int_0^1\varepsilon_1\,d\theta\,K(\lambda-\lambda_0)\,ds\Big|.
$$

For $x\in[0,h]$: $\Delta(x)\le\varepsilon_1 h|\lambda-\lambda_0|+\varepsilon_1 Kh|\lambda-\lambda_0|+K\int_0^x\Delta(s)\,ds$. By Gronwall, $\Delta(x)\le\varepsilon_1(K+1)h\,|\lambda-\lambda_0|\,e^{Kx}$, hence

$$
\frac{\Delta(x)}{|\lambda-\lambda_0|}\le\varepsilon_1(K+1)h\,e^{Kh}<\varepsilon\quad\Rightarrow\quad\lim_{\lambda\to\lambda_0}\frac{\Delta(x)}{|\lambda-\lambda_0|}=0.
$$

Similar proof for $x\in[-h,0]$. □

Cor. If $f\in C^k(\Omega)$, then the solution $\varphi(x,\lambda)$ is C^k on D (by induction).

Thm. If f is analytic in Ω (or analytic w.r.t. λ), then the solution is analytic in D (or analytic w.r.t. λ).

Eg. $\begin{cases}\frac{dy}{dx}=y+\lambda(x+y^2)\\ y(0)=1\end{cases}$: find $\frac{\partial y}{\partial\lambda}\big|_{\lambda=0}$.

Sol. $y(x,\lambda)=1+\int_0^x y(s,\lambda)+\lambda(s+y^2(s,\lambda))\,ds$, so

$$
\frac{\partial y}{\partial\lambda}(x,\lambda)=\int_0^x\frac{\partial y}{\partial\lambda}(s,\lambda)+s+y^2(s,\lambda)+2\lambda\,y(s,\lambda)\,\frac{\partial y}{\partial\lambda}(s,\lambda)\,ds;
$$

hence $u(x,\lambda):=\frac{\partial y}{\partial\lambda}(x,\lambda)$ solves

$$
\begin{cases}\frac{d}{dx}u(x,\lambda)=(1+2\lambda\,y(x,\lambda))\,u(x,\lambda)+(x+y^2(x,\lambda))\\ u(0,\lambda)=0.\end{cases}
$$

Solving the first order ODE: $e^{-\int_0^x A(s)\,ds}\,u(x,\lambda)=\int_0^x e^{-\int_0^z A(s)\,ds}\,B(z)\,dz$ with $A=1+2\lambda y$, $B=x+y^2$. Plug in $\lambda=0$:

$$
e^{-x}\,u(x,0)=\int_0^x(ze^{-z}+e^z)\,dz=e^x-e^{-x}(x+1),\qquad\text{hence}\quad u(x,0)=e^{2x}-(x+1).\qquad\square
$$

Chapter 3 Linear Systems

$y(x) : I \subseteq \mathbb{R} \to \mathbb{R}^n$, $\dfrac{dy}{dx} = A(x)y + f(x)$, $A, f \in C(a, b)$.

3.1 General theory

Lemma. The solution of the IVP is **unique & extends** to (a, b).

Pf. It suffices to prove $y(x)$ bounded on $(c, d) \subseteq (a, b)$:

$$\frac{d}{dx}\|y(x)\|^2 = y(x)^T\big(A(x)y + f(x)\big) + \big(y^T A^T + f^T\big)y(x) \le 2\,\|y(x)\|\,\big(\|A(x)\|\,\|y(x)\| + \|f(x)\|\big),$$

$$\Rightarrow\ \frac{d}{dx}\|y(x)\| \le \|A(x)\|\,\|y(x)\| + \|f(x)\| \le M\|y(x)\| + N.$$

By Gronwall, $\|y(x)\|$ is bounded. □

3.1.1 Homogeneous equation

Lemma. $S = \big\{y(x) \text{ solves } \frac{dy}{dx} = A(x)y\big\}$ is an n-dim vector space.

Pf. Obviously S is a vector space; check the dimension. Consider $T : \mathbb{R}^n \to S$, $y_0 \mapsto$ the solution $y(x)$ with $y(0) = y_0$. T is linear and bijective, making $S \cong \mathbb{R}^n$. So $\dim S = n$. □

Lemma. There are n linearly independent solutions $\{y_i(x)\}$ s.t. every $f \in S$ is $y = \sum_{i=1}^n a_i y_i$. We call $\{y_i(x)\}$ a fundamental system of solutions.

Pf. Using T, choose $y_0^{(1)}, \dots, y_0^{(n)}$ to get $y_i(x)$. □

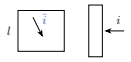
Def. For n solutions of the homogeneous equation, $W(y_1, \dots, y_n) := \det(y_1(x)\ \cdots\ y_n(x))$ is the **Wronskian determinant**.

Lemma ((Liouville identity))**.**

$$W(x) = W(x_0)\, e^{\int_{x_0}^x \operatorname{tr} A(s)\, ds}, \qquad \forall\ x_0, x \in (a, b).$$

Pf. We prove $W'(x) = \operatorname{tr} A(x)\, W(x)$:

$$W'(x) = \sum_{i=1}^n \begin{vmatrix} y_{11}(x) & y_{21}(x) & \cdots & y_{n1}(x) \\ \vdots & \vdots & & \vdots \\ y'_{1i}(x) & y'_{2i}(x) & \cdots & y'_{ni}(x) \\ \vdots & \vdots & & \vdots \\ y_{1n}(x) & y_{2n}(x) & \cdots & y_{nn}(x) \end{vmatrix}, \qquad y'_k(x) = A(x)\, y_k(x)\ \Rightarrow\ y'_{kl} = \sum_{m=1}^n a_{lm}\, y_{km}.$$



Thus

$$\begin{vmatrix} y_{11}(x) & y_{21}(x) & \cdots & y_{n1}(x) \\ \sum_m a_{im} y_{1m} & \sum_m a_{im} y_{2m} & \cdots & \sum_m a_{im} y_{nm} \\ \vdots & \vdots & & \vdots \\ y_{1n}(x) & y_{2n}(x) & \cdots & y_{nn}(x) \end{vmatrix} = a_{\overline{ii}}\, W(x)$$

(only if $m = i$ is the determinant nonzero). Hence $W'(x) = \operatorname{tr}(A(x))\, W(x)$, and $W(x) = W(x_0)\, e^{\int_{x_0}^x \operatorname{tr} A(s)\, ds}$. □

Lemma. $y_1(x), \dots, y_n(x) \in S$ are linearly dependent $\iff$ $W(x) \equiv 0$ on (a,b) $\iff$ $W(x_0) = 0$ for some $x_0 \in (a,b)$.

Pf. The first equivalence is the definition; the second follows from the Liouville identity. $\square$

Def. Suppose $y_1(x), \dots, y_n(x)$ are fundamental solutions of $y' = Ay$; denote $Y := (y_1 \;\; y_2 \;\; \cdots \;\; y_n)$ a **fundamental matrix**. If $Y(x_0) = I_n$, we call Y a principal fundamental matrix.

Lemma. (1) If $\Phi(x)$ is a fundamental matrix and $C \in M_n(\mathbb{R})$ is constant with $\det C \neq 0$, then $\Phi\, C$ is also a fundamental matrix.

(2) Any two fundamental matrices differ by a constant nonsingular matrix: $\forall\ \Phi, \Psi$, $\exists\ C$ with $\det C \neq 0$ & $\Psi = \Phi\, C$.

(3) Given any initial data $y(x_0)$, the solution is $y(x) = \Phi(x)\alpha$ with $y(x_0) = \Phi(x_0)\alpha$, $\alpha \in \mathbb{R}^n$ constant.

Pf. (1) $(\Phi C)' = \Phi' C = A\Phi C$, $\det(\Phi C) \neq 0$.

(2) Consider $\Phi^{-1}(x)\Psi(x)$:

$$\frac{d}{dx}\Big(\Phi^{-1}(x)\Psi(x)\Big) = \Big(-\Phi^{-1}\,\frac{d}{dx}\Phi\,\Phi^{-1}\Big)\Psi(x) + \Phi^{-1}\,\frac{d}{dx}\Psi = \big(-\Phi^{-1}A\Phi\Phi^{-1}\big)\Psi(x) + \Phi^{-1}A\Psi \equiv 0,$$

hence $\Phi^{-1}(x)\Psi(x) = C$ with $\det C \neq 0$.

(3) Take $\alpha \in \mathbb{R}^n$ s.t. $y(x_0) = \Phi(x_0)\alpha$; then $\Phi(x)\alpha$ solves the IVP. $\square$

3.1.2 Inhomogeneous equation $y' = A(x)y + f(x)$

Lemma. The general solution has the form $\varphi(x) = \Phi(x)\, C + \varphi^*(x)$, where $\Phi(x)$ is a fundamental matrix of $y' = A(x)y$ and $\varphi^*(x)$ is any solution of $y' = A(x)y + f(x)$.

Lemma. The general solution has the form

$$\varphi(x) = \Phi(x)\Big(C + \int_{x_0}^{x} \Phi^{-1}(s) f(s)\, ds\Big).$$

So we only need to solve the **homogeneous** equation.

d'Alembert reduction of order. For $y' = A(x)y$ with one nonzero solution $\varphi_1(x)$, set (W.L.O.G. $\varphi_{11}(x) \neq 0$)

$$H(x) := \begin{pmatrix} \varphi_{11}(x) & 0 & \cdots & 0 \\ \varphi_{12}(x) & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_{1n}(x) & 0 & \cdots & 1 \end{pmatrix}, \qquad y(x) = H(x)\, z(x).$$

Then $y'(x) = H'(x)z(x) + H(x)z'(x)$ and also $y'(x) = A(x)H(x)z(x)$, so

$$z'(x) = H^{-1}(x)\big(A(x)H(x) - H'(x)\big)z(x) = H^{-1}(x)\, A(x) \begin{pmatrix} 0 \\ I \end{pmatrix} z(x),$$

reducing the original problem to an $(n-1)$-order problem.

Eg. $\frac{dy}{dx} = A(x)y$, $A(x) = \begin{pmatrix} x^2 & -1 \\ 2x & 0 \end{pmatrix}$.

Sol. Observe $\varphi(x) = \begin{pmatrix} 1 \\ x^2 \end{pmatrix}$ is a solution. Take $H(x) := \begin{pmatrix} 1 & 0 \\ x^2 & 1 \end{pmatrix}$, $y(x) = H(x)z(x)$:

$$z'(x) = H^{-1}(x)A(x) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} z(x) \xrightarrow{\text{reduction}} \begin{cases} z_1'(x) = -z_2(x), \\ z_2'(x) = x^2 z_2(x). \end{cases}$$

3.2 Linear systems with constant coefficients

Consider $y' = Ay + f(x)$ with A a constant matrix; find a fundamental matrix of $y' = Ay$.

Def. Take $(M_n(\mathbb{R}), \|\cdot\|)$ as a complete metric space with $\|A\| := \sum_{1\le i,j\le n} |a_{ij}|$. Then $e^A := \sum_{k\ge 0} \frac{A^k}{k!}$ and $e^{Ax} := \sum_{k\ge 0} \frac{(Ax)^k}{k!}$.

Prop. If $[A,B] := AB - BA = 0$, then $e^{A+B} = e^A e^B$.

Pf.

$$e^{A+B} = \sum_{n=0}^{+\infty} \frac{(A+B)^n}{n!} = \sum_{n=0}^{+\infty} \frac{\sum_{k=0}^n \binom{n}{k} A^k B^{n-k}}{n!} = \sum_{n=0}^{+\infty}\sum_{k=0}^{n} \frac{A^k}{k!}\cdot\frac{B^{n-k}}{(n-k)!} = \Big(\sum_{k\ge 0}\frac{A^k}{k!}\Big)\Big(\sum_{l\ge 0}\frac{B^l}{l!}\Big),$$

(rigorously, using the sums of the first n terms). □

Prop. (1) $\forall A \in M_n(\mathbb{R})$, e^A is invertible & $(e^A)^{-1} = e^{-A}$.
(2) If $[A,B] = 0$, then $[e^A, B] = 0$.
(3) $A, P \in M_n(\mathbb{R})$ with P invertible $\Rightarrow e^{PAP^{-1}} = Pe^A P^{-1}$.

Thm. e^{Ax} is a fundamental matrix of $y' = Ay$.

Pf. e^{Ax} converges uniformly on any compact set, so we can differentiate term by term. □

Case 1. $A = \operatorname{diag}\{\lambda_1, \dots, \lambda_n\}$: then $e^{Ax} = \operatorname*{diag}_{\lambda\ \ 1}\{e^{\lambda_1 x}, \dots, e^{\lambda_n x}\}$.

Case 2. A is a Jordan form, i.e. $A = \begin{pmatrix} \ddots & \ddots & \\ & \ddots & \underset{\lambda}{1} \end{pmatrix} =: A_1 + N$. Notice $[A_1, N] = 0$, so we calculate e^{Nx}:

$$e^{Nx} = \sum_{k\ge 0}\frac{(Nx)^k}{k!} = I + \begin{pmatrix} 0 & x & & \\ & \ddots & \ddots & \\ & & \ddots & x \\ & & & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & \frac{x^2}{2} & \\ & \ddots & \ddots & \frac{x^2}{2} \\ & & \ddots & \\ & & & 0 \end{pmatrix} + \cdots + \begin{pmatrix} & & & \frac{x^{n-1}}{(n-1)!} \\ & & & \\ & & & \\ 0 & & & \end{pmatrix},$$

$$\text{thus} \quad e^{Ax} = \begin{pmatrix} e^{\lambda x} & xe^{\lambda x} & \cdots & \frac{x^{n-1}}{(n-1)!}e^{\lambda x} \\ & e^{\lambda x} & \cdots & \frac{x^{n-2}}{(n-2)!}e^{\lambda x} \\ & & \ddots & \vdots \\ & & & e^{\lambda x} \end{pmatrix}.$$

Case 3. $A = \operatorname{diag}\{J_1, \dots, J_k\}$ with J_i Jordan blocks: $e^{Ax} = \operatorname{diag}\{e^{J_1 x}, \dots, e^{J_k x}\}$.
Case 4. Any given A with $A = PBP^{-1}$: $e^{Ax} = Pe^{Bx}P^{-1}$.

Further simplification. Case 1. A has n simple eigenvalues $\lambda_1, \dots, \lambda_n$ with eigenvectors $\vec{\xi}_i$: then $e^{Ax}P = (e^{\lambda_1 x}\vec{\xi}_1,\ \dots,\ e^{\lambda_n x}\vec{\xi}_n)$ is already a fundamental matrix. Here the matrix can be complex; taking $e^{\lambda_i x}\vec{\xi}_i + e^{\bar{\lambda}_i x}\bar{\vec{\xi}_i}$ gives a real solution.

Case 2. A has repeated eigenvalues $\lambda_1, \dots, \lambda_s$, where λ_j has multiplicity n_j; then

$$\varphi(x) = e^{\lambda_j x}\Big(\vec{\xi}_0 + x\vec{\xi}_1 + \cdots + \frac{x^{n_j-1}}{(n_j-1)!}\vec{\xi}_{n_j-1}\Big)$$

is a solution of $y' = Ay$ iff $(A-\lambda_j I)^{n_j}\vec{\xi}_0 = 0$ with $\vec{\xi}_1 = (A-\lambda_j I)\vec{\xi}_0$, $\vec{\xi}_2 = (A-\lambda_j I)^2\vec{\xi}_0$, $\dots$, $\vec{\xi}_{n_j-1} = (A-\lambda_j I)^{n_j-1}\vec{\xi}_0$. So we have the fundamental matrix

$$\Phi(x) = \big(e^{\lambda_1 x}P_1^{(1)}(x),\ e^{\lambda_1 x}P_2^{(1)}(x),\ \cdots,\ e^{\lambda_1 x}P_{n_1}^{(1)}(x),\ \cdots,\ e^{\lambda_j x}P_1^{(j)}(x),\ \cdots,\ e^{\lambda_s x}P_{n_s}^{(s)}(x)\big),$$

where

$$P_j^{(i)}(x) = \vec{\xi}_{j0}^{(i)} + x\vec{\xi}_{j1}^{(i)} + \cdots + \frac{x^{n_j-1}}{(n_j-1)!}\vec{\xi}_{j,n_j-1}^{(i)},$$

with $\vec{\xi}_{1,0}^{(i)}, \ldots, \vec{\xi}_{n_j,0}^{(i)}$ independent solutions of $(A-\lambda_j I)^{n_j}\vec{\xi} = 0$.

(Margin steps: (1) $A \to f_A(x)$, λ_i with n_i; (2) $\mathrm{Ker}(A-\lambda_i I)^{n_i} \to \vec{\xi}_{i,1}, \ldots, \vec{\xi}_{i,n_i}$ — get the generating space $(A-\lambda_i I)^m \vec{\xi}_{i,1}$, $m \le n_i - 1$; (3) form solutions.)

Rem. (1) $\Phi(x)$ could be a complex matrix. To construct a real fundamental matrix, take $e^{Ax} = \Phi(x)\cdot\Phi^{-1}(0)$ (a constant matrix times Φ).

(2) $\mathrm{Re}(\lambda_i) < 0$ for every eigenvalue λ_i of $A \iff$ every solution of $y' = Ay$ converges to 0 as $x \to +\infty$.

3.3 Higher order ODE

3.3.1 General theory

Consider

$$y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n(x)y = f(x). \qquad (*)$$

Denote

$$Y(x) = \begin{pmatrix} y(x) \\ \vdots \\ y^{(n-1)}(x) \end{pmatrix}, \quad A(x) = \begin{pmatrix} 0 & 1 & & & 0 \\ 0 & 0 & 1 & & \\ \vdots & & \ddots & \ddots & \\ 0 & \cdots & & 0 & 1 \\ -a_n(x) & -a_{n-1}(x) & \cdots & & -a_1(x) \end{pmatrix}, \quad F(x) := \begin{pmatrix} 0 \\ \vdots \\ 0 \\ f \end{pmatrix};$$

then $(*) \iff Y' = AY + F$. So **existence, extension, linear independence** are the same.

Thm. (1) The IVP for $(*)$ has a unique solution which extends to (a,b) if the $a_j(x)$ are defined on (a,b).

(2) For the homogeneous equation ($f(x) \equiv 0$) the solutions form an n-dim linear space.

(Here we have the Wronskian determinant and the Liouville identity, as in Section 3.1.1.)

Thm. Given arbitrary $\varphi_1, \ldots, \varphi_n$ linearly independent on (a,b) with $\varphi_j \in C^\infty(a,b)$, we can find a **unique** n-th order homogeneous ODE such that $\{\varphi_j\}$ is a fundamental system of solutions for it.

Pf. Suppose $\{\varphi_j\}$ satisfies $y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n(x)y = 0$, i.e.

$$\begin{pmatrix} \varphi_1^{(n-1)} & \cdots & \varphi_n^{(n-1)} \\ \vdots & & \vdots \\ \varphi_1 & \cdots & \varphi_n \end{pmatrix} \begin{pmatrix} a_1(x) \\ \vdots \\ a_n(x) \end{pmatrix} = -\begin{pmatrix} \varphi_1^{(n)} \\ \vdots \\ \varphi_n^{(n)} \end{pmatrix},$$

so we solve for $a_i(x)$. □

Prop (starred)**.** Consider $n = 2$: $y'' + p(x)y' + q(x)y = 0$, $p, q \in C(a,b)$. Given a solution $\varphi(x) \neq 0$, the general solution is

$$y = c_1\varphi(x) + c_2\varphi(x)\int_{x_0}^{x} \frac{1}{\varphi^2(s)}\, e^{-\int_{x_0}^{s} p(t)dt}\, ds.$$

Pf. Suppose $\psi(x)$ is another independent solution:

$$W(x) = \begin{vmatrix} \varphi & \psi \\ \varphi' & \psi' \end{vmatrix} = W(x_0)\exp\Big(\int_{x_0}^{x} -p(s)\, ds\Big),$$

so $\big(\frac{\psi}{\varphi}\big)' = \frac{1}{\varphi^2}W(x_0)\exp\big(\int_{x_0}^{x} -p(s)\, ds\big)$, hence $\frac{\psi}{\varphi} = C\int_{x_0}^{x} \frac{W(x_0)}{\varphi(s)^2}\exp\big(\int_{x_0}^{s} -p(t)dt\big)\, ds$. (Margin: the Wronski method; useful in power series.) □

Prop. Given fundamental solutions $\varphi_1, \dots, \varphi_n$ of $y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n(x)y = 0$, the general solution of the inhomogeneous equation $y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n y = f(x)$ is

$$y(x) = \sum_{i=1}^{n} c_i \varphi_i(x) + \varphi^*(x),$$

where $\varphi^*(x)$ is a special solution.

Pf. For $Y' = AY + F$ we proved $Y(x) = \Phi(x)\big(C + \int_{x_0}^{x} \Phi^{-1}(s)F(s)\,ds\big)$; the first component gives the claim. □

Alternate method (variation of parameters). Guess $y = \sum c_j(x)\varphi_j(x)$:

$$\begin{aligned}
y'(x) &= \sum c_j'(x)\varphi_j(x) + \sum c_j(x)\varphi_j'(x) && \text{suppose } \textstyle\sum c_j'(x)\varphi_j(x) = 0,\\
y''(x) &= \sum c_j'(x)\varphi_j'(x) + \sum c_j(x)\varphi_j''(x) && \text{suppose } \textstyle\sum c_j'(x)\varphi_j'(x) = 0,\\
&\vdots\\
y^{(n-1)}(x) &= \sum c_j'(x)\varphi_j^{(n-2)}(x) + \sum c_j(x)\varphi_j^{(n-1)}(x) && \text{suppose } \textstyle\sum c_j'(x)\varphi_j^{(n-2)}(x) = 0,\\
y^{(n)}(x) &= \sum c_j'(x)\varphi_j^{(n-1)}(x) + \sum c_j(x)\varphi_j^{(n)}(x).
\end{aligned}$$

Substituting into $y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n(x)y = f(x)$ gives $\sum c_j'(x)\varphi_j^{(n-1)}(x) = f(x)$. So

$$\begin{pmatrix} \varphi_1 & \cdots & \varphi_n \\ \vdots & & \vdots \\ \varphi_1^{(n-1)} & \cdots & \varphi_n^{(n-1)} \end{pmatrix} \begin{pmatrix} c_1' \\ \vdots \\ c_n' \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ f(x) \end{pmatrix} \Rightarrow \vec{C}' = \Phi^{-1}(x)\,(0, \dots, 0, f)^T = \frac{1}{W(x)} \begin{pmatrix} \hat{\Phi}_1 \\ \vdots \\ \hat{\Phi}_n \end{pmatrix},$$

where $\hat{\Phi}_j$ is the determinant with the j-th column replaced by $(0, \dots, 0, f)^T$: $\hat{\Phi}_j(x) = f(x)W_j(x)$. Hence

$$c_j = \int_{x_0}^{x} \frac{\hat{\Phi}_j(s)}{W(s)}\,ds = \int_{x_0}^{x} \frac{W_j(s)f(s)}{W(s)}\,ds,$$

giving a special solution.

3.3.2 n-th order ODE with constant coefficients

Consider $y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y = 0$, $a_i \in \mathbb{R}$. Denote $p(\lambda) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_n$ and $L = \frac{d^n}{dx^n} + a_1\frac{d^{n-1}}{dx^{n-1}} + \cdots + a_n$.

Lemma. $L(e^{\lambda x}) = p(\lambda)e^{\lambda x}$ for all λ.

Lemma. If λ_j is a root of $p(\lambda)$ with multiplicity n_j, then $L(x^{k_j}e^{\lambda_j x}) = 0$ for $k_j = 0, \dots, n_j - 1$. (Simple computation.)

So we only need to solve $p(\lambda) = 0$ to get all solutions.

Thm. Suppose $p(\lambda) = 0$ has distinct roots $\lambda_1, \dots, \lambda_s$ with multiplicities $n_1, \dots, n_s$ ($n_1 + \cdots + n_s = n$). Then $Ly = 0$ has n linearly independent solutions $x^{k_j}e^{\lambda_j x}$, $k_j = 0, \dots, n_j - 1$, $1 \le j \le s$.

Pf. We only need linear independence. Suppose $\sum_{j=1}^{s}\sum_{k_j=0}^{n_j-1} c_{j,k_j}\,x^{k_j}e^{\lambda_j x} \equiv 0$ on (a, b) with not all $c_{j,k_j} = 0$; i.e. $\sum_{j=1}^{s} P_j(x)e^{\lambda_j x} \equiv 0$, where $\deg P_j \le n_j - 1$. W.L.O.G. $P_1 \not\equiv 0$:

$$P_1(x) + \sum_{j=2}^{s} P_j(x)e^{(\lambda_j - \lambda_1)x} \equiv 0 \xrightarrow{\frac{d^{n_1}}{dx^{n_1}}} \sum_{j\ge 2} Q_j(x)e^{(\lambda_j - \lambda_1)x} \equiv 0$$

with $Q_s(x) = c(\lambda_s - \lambda_1)^{n_s}x^k + \cdots$. Then $Q_2(x) + \sum_{j\ge 3} Q_j(x)e^{(\lambda_j - \lambda_2)x} \equiv 0 \xrightarrow{\frac{d^{n_2}}{dx^{n_2}}} \sum_{j\ge 3} \widetilde{Q}_j(x)e^{(\lambda_j - \lambda_2)x} \equiv 0$. Repeating, we get $R(x)e^{(\lambda_s - \lambda_{s-1})x} \equiv 0$: contradiction. □

Rem. If $p(\lambda)=0$ has a complex root λ_0, take $\mathrm{Re}(x^k e^{\lambda_0 x})$ and $\mathrm{Im}(x^k e^{\lambda_0 x})$ to obtain real solutions.

Eg. $y''(x)+\alpha^2 y(x)=f(x)$, $y(0)=y_0$, $y'(0)=y_1$.

$p(\lambda)=\lambda^2+\alpha^2=0 \Rightarrow \lambda=\pm\alpha i$, and $e^{i\alpha x}=\cos\alpha x+i\sin\alpha x$, so $[\cos\alpha x \;\; \sin\alpha x]$ is a fundamental system of the homogeneous equation. $W=\alpha$; $W_1=-\sin\alpha x$, $W_2=\cos\alpha x$. The general solution is

$$y=C_1\cos\alpha x+C_2\sin\alpha x+\cos\alpha x\int_0^x\frac{W_1(s)}{W(s)}f(s)\,ds+\sin\alpha x\int_0^x\frac{W_2(s)}{W(s)}f(s)\,ds.$$

Eg (PDE application)**.** $u(t,x):\mathbb{R}\times\mathbb{R}^n\to\mathbb{R}$ with $(\partial_t^2-\Delta_x)u=f(t,x)$, $u(0,x)=u_0(x)$, $\partial_t u(0,x)=u_1(x)$. With $\hat{h}(\xi)=\int h(x)e^{-ix\cdot\xi}dx$ and inverse $\check{g}(x)=\frac{1}{(2\pi)^n}\int g(\xi)e^{ix\cdot\xi}d\xi$, applying the Fourier transform:

$$\hat{u}_{tt}+|\xi|^2\hat{u}=\hat{f},\qquad \hat{u}(t,\xi)=\hat{u}_0(\xi)\cos t|\xi|+\frac{\hat{u}_1(\xi)}{|\xi|}\sin t|\xi|+\int_0^t\frac{\sin((t-s)|\xi|)}{|\xi|}\hat{f}(s,\xi)\,ds.$$

Taking $\mathcal{F}^{-1}$:

$$u(t,x)=\cos(t|\nabla|)\,u_0+\frac{\sin(t|\nabla|)}{|\nabla|}\,u_1+\int_0^t\frac{\sin((t-s)|\nabla|)}{|\nabla|}\,f(s,x)\,ds.$$

3.3.3 Special cases of the inhomogeneous equation

Case 1. $Ly=P_m(x)e^{\mu x}$.

(a) If $p(\mu)\neq 0$: guess $Q_m(x)e^{\mu x}$ and determine Q_m.

(b) If μ is a root of $p(\lambda)$ with multiplicity k: recall $L(x^l e^{\lambda x})=\sum_j\binom{l}{j}p^{(j)}(\lambda)x^{l-j}e^{\lambda x}$; putting $p(\mu)=\cdots=p^{(k-1)}(\mu)=0$,

$$L(x^l e^{\mu x})=\sum_{j\geq k}\binom{l}{j}p^{(j)}(\mu)x^{l-j}e^{\mu x}=e^{\mu x}Q_{l-k}(x),$$

so we guess $x^k Q_m(x)e^{\mu x}$.

Case 2. $Ly=e^{\alpha x}(A_m(x)\sin\beta x+B_l(x)\cos\beta x)$: guess $x^k e^{\alpha x}(C_N(x)\sin\beta x+D_N(x)\cos\beta x)$, where k is the multiplicity of $\mu=\alpha+i\beta$ as a root of $p(\lambda)=0$, and $N=\max\{m,l\}$.

Eg. $y''+y=xe^x$: $p(\lambda)=0\Rightarrow\lambda=\pm i$, so $\{\cos x,\sin x\}$ is a fundamental system. Guess $y=(Ax+B)e^x$: $A=\frac{1}{2}$, $B=-\frac{1}{2}$, so $y=\frac{1}{2}(x-1)e^x+C_1\cos x+C_2\sin x$. □

Eg ((Euler equation), starred)**.** $x^n y^{(n)}+a_1x^{n-1}y^{(n-1)}+\cdots+a_{n-1}xy'+a_n y=0$.

Method (1). Set $x=e^t$, $\tilde{y}(t)=y(e^t)$: $\frac{d\tilde{y}}{dt}=xy'$, $\frac{d^2\tilde{y}}{dt^2}=x^2y''+xy'$; we get an equation with constant coefficients.

Method (2). Try x^λ. Denote $D_n:=x^n\frac{d^n}{dx^n}+a_1x^{n-1}\frac{d^{n-1}}{dx^{n-1}}+\cdots+a_n$. Since $x^k(x^\lambda)^{(k)}=\lambda(\lambda-1)\cdots(\lambda-k+1)x^\lambda$,

$$D_n(x^\lambda)=\big[\lambda(\lambda-1)\cdots(\lambda-n+1)+a_1\lambda(\lambda-1)\cdots(\lambda-n+2)+\cdots+a_n\big]x^\lambda=:q(\lambda)x^\lambda.$$

Hence if $q(\lambda_0)=0$, then x^{λ_0} is a solution. If λ_0 has multiplicity k: $D_n(x^\lambda(\ln x)^l)=\sum_j\binom{l}{j}q^{(j)}(\lambda)(\ln x)^{l-j}x^\lambda$, so $D_n(x^{\lambda_0}(\ln x)^l)\equiv 0$ for $l\leq k-1$. We get solutions

$$x^{\lambda_j}(\ln x)^{k_j},\qquad 0\leq k_j\leq n_j-1,\quad 1\leq j\leq s,$$

where $\lambda_1,\ldots,\lambda_s$ are the distinct roots of $q(\lambda)$ with multiplicities n_j. □

3.4 Linear systems with periodic coefficients (Floquet theory)

Consider $y' = A(x)y$, $y : \mathbb{R} \to \mathbb{R}^n$, $A(x+w) = A(x)$, $w > 0$.

Thm. (1) If $\Phi(x)$ is a fundamental matrix with $\Phi(0) = I$, then $\Phi(x+w)$ is also a fundamental matrix, and $\exists$ a constant nonsingular CC s.t. $\Phi(x+w) = \Phi(x)C$ (indeed $C = \Phi(w)$).

(2) For $w \neq 0$ and CC nonsingular: (a) $\exists$ R (possibly complex) s.t. $e^{wR} = C$; (b) $\exists$ $\widetilde{R}$ real s.t. $e^{w\widetilde{R}} = C^2$.

(3) With $P(x) = \Phi(x)e^{-xR}$, $P(x)$ is w-periodic; taking $y(x) = P(x)z(x)$ gives $z'(x) = Rz(x)$.

(4) With $\widetilde{P}(x) = \Phi(x)e^{-x\widetilde{R}}$ & $y(x) = \widetilde{P}(x)\widetilde{z}(x)$: $\widetilde{P}(x)$ is $2w$-periodic & $\widetilde{z}'(x) = \widetilde{R}\widetilde{z}(x)$.

Pf. We only prove (2); the rest is checked by computing. (a) Reduce to the Jordan form $J = \lambda I + N$:

$$J = \lambda I\Big(I + \frac{N}{\lambda}\Big), \qquad \log J := \log\lambda\, I + \log\Big(I + \frac{N}{\lambda}\Big) = \operatorname{diag}(\log\lambda, \dots, \log\lambda) + \sum_{n\geq 1} \frac{(-1)^{n+1}}{n}\Big(\frac{N}{\lambda}\Big)^n$$

(N nilpotent, so the sum is finite); verify $e^{\log J} = J$ by comparing coefficients. (b) $\lambda > 0 \Rightarrow \log\lambda$ real $\Rightarrow \log J$ real.

If λ is an eigenvalue of CC, then λ^2 is an eigenvalue of C^2. Let C^2 have Jordan form $\operatorname{diag}\{J_1, \dots, J_k\}$; J is either $\begin{pmatrix} \lambda & 1 & \\ & \ddots & \ddots \\ & & \lambda \end{pmatrix}$ or a block $\begin{pmatrix} T^E & & \\ T & T^E & \\ & \ddots & \ddots \end{pmatrix}$ with $E = I_{2\times 2}$, $T = r\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, $r > 0$:

$$Q = \log T = \begin{pmatrix} \log r & \\ & \log r \end{pmatrix} + \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix},$$

$$\log J = \begin{pmatrix} Q & & \\ & \ddots & \\ & & Q \end{pmatrix} + \sum_{h\geq 1} \frac{(-1)^{n+1}}{n} \begin{pmatrix} 0 & T^{-1} & \\ & \ddots & \ddots \\ & & T^{-1} \end{pmatrix}^h.$$

Real! □

Rem. R or $\widetilde{R}$ are not unique: $C = e^{wR} = e^{w(R + \frac{2k\pi i}{w} I)}$.

Def. An eigenvalue of CC is called a **characteristic multiplier** of A; an eigenvalue of R is called a **characteristic exponent** (NOT unique).

Thm. For $y' = A(x)y$ and any characteristic multiplier λ, there is a solution (possibly complex) s.t. $y(x+w) = \lambda y(x)$. Equivalently, taking the characteristic exponent μ with $e^{\mu w} = \lambda$, there is a solution $y(x) = h(x)e^{\mu x}$ with $h(x)$ periodic.

Pf. Take an eigenvector y_0 of R for μ; then $y(x) = \Phi(x)y_0$ is a solution. Using $\Phi(x) = P(x)e^{xR}$ and $\Phi(x+w)y_0 = \Phi(x)Cy_0 = \lambda\Phi(x)y_0$:

$$y(x) = P(x)e^{wR}y_0 = P(x)e^{w\mu}y_0 = e^{w\mu}P(x)y_0.$$

With $h(x) = P(x)y_0$, which is w-periodic, we are done. □

Cor. $y' = A(x)y$ has a periodic solution $\iff$ 1 is a characteristic multiplier.

Pf. ($\Leftarrow$) by the periodic thm. ($\Rightarrow$) if $y(x)$ is a w-periodic solution, then $y(x) = \Phi(x)y_0$ and $y(x+w) = \Phi(x+w)y_0 = \Phi(x)Cy_0 = \Phi(x)y_0$, so $Cy_0 = y_0$, i.e. 1 is a characteristic multiplier. □

Cor. (1) If all characteristic multipliers satisfy $|\lambda| < 1$ (equivalently $\operatorname{Re}\mu < 0$), then all solutions converge to 0 as $x \to +\infty$; i.e. the 0 solution is **asymptotically stable**.

Pf. $y(x) = P(x)z(x)$ with P bounded (w-periodic); $z'(x) = Rz(x)$ gives $z(x) = \sum e^{\mu_i x}P_{i,n_i}(x)$, and $\operatorname{Re}\mu_i < 0 \Rightarrow z(x) \to 0 \Rightarrow y(x) \to 0$. □

(2) If there exists $|\lambda| > 1$, then 0 is asymptotically unstable.

(3) If all $|\lambda| \leq 1$ and algebraic multiplicity = geometric multiplicity, then 0 is orbitally stable.

Chapter 4 Method of power series

4.1 Cauchy theory

Consider $\begin{cases} y' = f(x,y), \\ y_0 = y(x_0), \end{cases}$ where $f(x,y)$ is **analytic** in the domain $\Omega \subseteq \mathbb{R}\times\mathbb{R}^n$. The Picard thm. gives existence & uniqueness; we want to show the solution is analytic.

Def. (I) $\sum_{\bar i,\bar j} a_{\bar i\bar j}(x-x_0)^{\bar i}(y-y_0)^{\bar j}$ and (II) $\sum_{\bar i,\bar j} A_{\bar i\bar j}(x-x_0)^{\bar i}(y-y_0)^{\bar j}$. If $|a_{\bar i\bar j}| \le |A_{\bar i\bar j}|$ and (II) converges, we call (II) a **majorant** of (I).

Lemma. If $f(x,y)$ is analytic in Ω, then $\exists\ M>0$ s.t.

$$F(x,y) := \frac{M}{\big(1-\frac{x-x_0}{a}\big)\big(1-\frac{y_1-y_{01}}{b}\big)\cdots\big(1-\frac{y_n-y_{0n}}{b}\big)}$$

is a majorant of $f(x,y)$ in $\big\{(x,y): |x-x_0|<a<\alpha,\ |y-y_0|<b<\beta\big\}$.

Pf. Write $f(x,y)=\sum_{\bar i,\bar j} a_{\bar i\bar j}(x-x_0)^{\bar i}(y-y_0)^{\bar j}$, convergent in Ω; then $\exists M>0$ with $|a_{\bar i\bar j}a^{\bar i}b^{\bar j}|\le M$ for all $\bar i,\bar j$. Denote $A_{\bar i,\bar j}=\frac{M}{a^{\bar i}b^{\bar j}}$:

$$\sum_{\bar i\bar j} A_{\bar i,\bar j}(x-x_0)^{\bar i}(y-y_0)^{\bar j} = M\sum_{\bar i,\bar j_k}\Big(\frac{x-x_0}{a}\Big)^{\bar i}\Big(\frac{y_1-y_{01}}{b}\Big)^{\bar j_1}\cdots\Big(\frac{y_n-y_{0n}}{b}\Big)^{\bar j_n} =: F(x,y),$$

with $|a_{\bar i\bar j}|\le|A_{\bar i\bar j}|$. □

Lemma. The system $\frac{dy_k}{dx}=F(x,y)$ (all k), $y(x_0)=y_0$, has an analytic solution on $|x-x_0|<\rho$, where $\rho = a\big(1-e^{-\frac{b}{aM(n+1)}}\big)$.

Pf. $\frac{dy_k}{dx}=\frac{dy_l}{dx}\Rightarrow y_k-y_{k_0}=y_l-y_{l_0}$; thus y_k $(k=1,\dots,n)$ satisfies

$$\frac{dz}{dx}=\frac{M}{\big(1-\frac{x-x_0}{a}\big)\big(1-\frac{z-z_0}{b}\big)^n},$$

$$\frac{b}{n+1}\Big(1-\frac{u-u_0}{b}\Big)^{n+1}-\frac{b}{n+1}=aM\ln\Big(1-\frac{x-x_0}{a}\Big),\qquad u=u_0+b-b\Big[1+\frac{(n+1)aM}{b}\ln\Big(1-\frac{x-x_0}{a}\Big)\Big]^{\frac{1}{n+1}}.$$

Requiring $\Big|\frac{(n+1)aM}{b}\ln\big(1-\frac{x-x_0}{a}\big)\Big|<1$ gives $|x-x_0|<\rho$. Then this solution can be written as a power series. □

Thm ((Cauchy))**.** For the IVP: if $f(x,y)$ is analytic in $\{|x-x_0|<\alpha,\ |y-y_0|<\beta\}$, then the IVP has a unique solution, analytic in $\{|x-x_0|<\rho\}$.

Pf. Uniqueness is Picard's thm. Write $f_k(x,y)=\sum_{\bar i,\bar j} a^k_{\bar i\bar j}(x-x_0)^{\bar i}(y-y_0)^{\bar j}$ and seek $y_k(x)=y_{0,k}+\sum_{m\ge1}c_{k,m}(x-x_0)^m$; we need (1) the $c_{k,m}$ to satisfy the equation, and (2) convergence of $y_k(x)$.

(1) By computation: $c_{k,1}=y_k'(x_0)=f_k(x_0,y_0)=a^k_{0,0}$, $c_{k,2}=\frac{1}{2!}y_k''(x_0)=\frac{1}{2!}(f'_{kx}+f'_{ky}+y')$; generally

$$c_{k,m}=\frac{1}{m!}y_k^{(m)}(x_0)=P_{k,m}\big(a^l_{\bar i\bar j}\big)\quad(1\le l\le n,\ \bar i+\bar j\le m-1),$$

depending only on $\bar i,\bar j,l$.

(2) Use $F_k(x,y)$: the analytic solution from the lemma is a majorant of $\sum c_{k,m}(x-x_0)^m$, so the latter converges. □

Eg ((Riccati equation)). $y' + y^2 = x$. Consider $y = \sum_{n\geq 0} c_n x^n$:

$$\sum_{k\geq 1} k c_k x^{k-1} + \sum_{m+n=k} c_m c_n x^k = x \quad \Rightarrow \quad \begin{cases} c_1 + c_0^2 = 0 \\ 2c_2 + c_0 c_1 + c_1 c_0 = 1 \\ (k+1)c_{k+1} + \sum_{m+n=k} c_m c_n = 0, \end{cases}$$

a recursion formula for the solution. □

4.2 Solving ODEs using power series

Consider $A(x)y'' + B(x)y' + C(x)y = 0$ with A, B, C analytic.

Easy case. If $\exists\, x_0$ with $A(x_0) \neq 0$, then $\exists r > 0$: $|x - x_0| < r \Rightarrow A(x) \neq 0$. Focus on $y'' + p(x)y' + q(x)y = 0$ with p, q analytic.

Thm. If $p(x), q(x)$ are analytic in $|x - x_0| < r$, then the solution is unique and analytic in $|x - x_0| < r$.

Pf. $\begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix}$, i.e. $w' = f(x, w)$ with f analytic in $|x - x_0| < r$, $|w - w_0| < \beta$ for every $\beta \leq +\infty$; so $\rho = r\big(1 - e^{-\frac{\beta}{3Mr}}\big) \to r$ as $\beta \to +\infty$. □

Eg ((Hermite eq.)). $y'' - 2xy' + \lambda y = 0$: with $y = \sum_{n\geq 0} c_n x^n$,

$$\sum_{k\geq 0} (k+2)(k+1)c_{k+2}x^k - 2x \sum_{k\geq 1} k c_k x^{k-1} + \lambda \sum_{k\geq 0} c_k x^k = 0,$$

$$n(n-1)c_n - 2(n-2)c_{n-2} + \lambda c_{n-2} = 0 \quad \Rightarrow \quad c_{n+2} = \frac{2n - \lambda}{(n+2)(n+1)} c_n.$$

If $\lambda \in 2\mathbb{Z}$, the solution is a polynomial, called (a) Hermite polynomial.

Eg ((Legendre eq.)). $(1 - x^2)y'' - 2xy' + \lambda(\lambda + 1)y = 0$ near $x = 0$: with $y = \sum_{n\geq 0} c_n x^n$, $y' = \sum_{n\geq 1} n c_n x^{n-1}$, $y'' = \sum_{n\geq 2} n(n-1)c_n x^{n-2}$,

$$(k+2)(k+1)c_{k+2} + (\lambda + k + 1)(\lambda - k)c_k = 0.$$

Complicated case. $A(x_0) = 0$: find solutions near x_0.

Eg. $x^2 y'' - 2y = 0$ near $x = 0$: the Euler equation gives $y = c_1 x^2 + \frac{c_2}{x}$ — not analytic. (Trying x^λ: $\lambda(\lambda - 1) - 2 = 0$, $\lambda = 2$ or $\lambda = -1$.)

Eg. $x^2 y'' + xy' + (x^2 - \frac{1}{4})y = 0$: denoting $u(x) = y\sqrt{x}$ we get $u'' + u = 0$, so the fundamental solutions are

$$y_1 = \frac{\cos x}{\sqrt{x}} = \sum_{k\geq 0} \frac{(-1)^k}{(2k)!} x^{2k - \frac{1}{2}}, \qquad y_2 = \frac{\sin x}{\sqrt{x}} = \sum_{k\geq 0} \frac{(-1)^k}{(2k+1)!} x^{2k + \frac{1}{2}}.$$

Def. If $f(x) = \sum_{n\geq 0} a_n (x - x_0)^{n+\rho}$, we call it a **general power series with index** ρ.

Thm ((Frobenius theory)). For $A(x)y'' + B(x)y' + C(x)y = 0$ (A, B, C analytic): if it can be written as

$$(x - x_0)^2 P(x)y'' + (x - x_0)Q(x)y' + R(x)y = 0$$

with $P(x_0) \neq 0$, at least one of $Q(x_0)$, $R(x_0)$ nonzero, and P, Q, R analytic in $|x - x_0| < r$, then x_0 is called a **regular singularity** for $A(x)$ (or for the equation).

Reduced to $(x - x_0)^2 y'' + p(x)(x - x_0)y' + q(x)y = 0$, the equation has a solution of the form

$$y = \sum_{n\geq 0} c_n (x - x_0)^{n+\rho}, \qquad c_0 \neq 0.$$

Rem. Recall the Euler equation, where $\frac{Q}{P} \equiv p_0$, $\frac{R}{P} \equiv q_0$: we get $x^2y'' + p_0xy' + q_0y$; guess $y = x^\rho$:

$$\big(\rho(\rho-1) + p_0\rho + q_0\big)x^\rho = 0 \;\Rightarrow\; \rho(\rho-1) + p_0\rho + q_0 = 0.$$

Pf. Suppose $x_0 = 0$, $p(x) = \sum_{n\ge0} p_n x^n$, $q(x) = \sum_{n\ge0} q_n x^n$. Set $y(x,\rho) = \sum_{n\ge0} c_n(\rho)x^{n+\rho}$ with $c_0(\rho) = 1$, and $L := x^2\frac{d^2}{dx^2} + xp(x)\frac{d}{dx} + q(x)$. Then

$$Ly(x,\rho) = \sum_{n\ge0} x^{n+\rho}\Big[c_n(\rho)\big((n+\rho)(n+\rho-1) + p_0(n+\rho) + q_0\big) + \sum_{l=0}^{n-1} c_l(\rho)\,p_{n-l}\,(l+\rho) + q_{n-l}\Big].$$

Writing the coefficient of $x^{n+\rho}$ as $c_nf_0(\rho+n) + c_{n-1}f_1(\rho+n-1) + \cdots + c_0f_n(\rho)$, where $f_0(\rho) := \rho(\rho-1) + p_0\rho + q_0$ and $f_j(\rho) := p_j\rho + q_j$: since $c_0(\rho) \ne 0$, we need $f_0(\rho) = 0$, which has either 2 real roots $\rho_1 \ge \rho_2$ or 2 complex roots. If $f_0(n+\rho) \ne 0$ for all $n \ge 1$, every c_n is determined. To find 2 solutions:

- **Case 1.** 2 complex roots: use any ρ to get c_n.
- **Case 2.** 2 real roots $\rho_1 > \rho_2$, $\rho_1 - \rho_2 \notin \mathbb{Z}$: use any ρ.
- **Case 3.** $\rho_1 = \rho_2$, or 2 real roots with $\rho_1 - \rho_2 \in \mathbb{Z}$: use ρ_1 — only one solution is found.

It remains to prove convergence of $\sum_{n\ge0} c_n(x-x_0)^n$. Suppose $\sum p_n(x-x_0)^n$, $\sum q_n(x-x_0)^n$ converge on $0 < x < r$. For $0 < r_i < r$ there is $M > 1$ with $|p_nr_i^n| < M$, $|q_nr_i^n| < M$ for all n. Now $|c_0(\rho)| = 1$, $|f_0(n+\rho)| \ge n^2$ (computing), $|c_1(\rho)| = \frac{|p_1\rho+q_1|}{|f_0(1+\rho)|}|c_0| \le \frac{M}{r_1}$. Guess $|c_n(\rho)| \le \big(\frac{M}{r_1}\big)^n$; induction proves it. So the solution converges in $|x| < \frac{r_1}{M}$. □

Rem. If $\rho_1 = \rho_2$ is repeated, the process above gives only 1 solution. With $y_1 = \sum_{n\ge0} c_n(x-x_0)^n$, another solution is

$$y_2 = y_1(x)\ln x + \sum_{n\ge0} b_n x^{n+\rho},$$

with computable b_n.

Pf. For most ρ with $f_0(n+\rho) \ne 0$ we can solve $c_n(\rho)$ from $c_0(\rho) = 1$, $c_nf_0(\rho+n) + c_{n-1}f_1(\rho+n-1) + \cdots + c_0f_n(\rho) = 0$; then $Ly(x,\rho) = x^\rho f_0(\rho)$ and

$$L\frac{\partial}{\partial\rho}y(x,\rho) = x^\rho f_0'(\rho) + x^\rho f_0(\rho)\ln x.$$

At $\rho = \rho_1$: $L\frac{\partial}{\partial\rho}y(x,\rho)\big|_{\rho=\rho_1} = 0$. So another solution is $y_2 = \frac{\partial}{\partial\rho}y(x,\rho)\big|_{\rho=\rho_1} = y_1(x)\ln x + \sum_{n\ge0} b_n x^{n+\rho}$ (using $y(x,\rho) = \sum_{n\ge0} c_n(x-x_0)^{n+\rho}$). □

Rem. If $f_0(\rho) = 0$ has 2 real solutions $\rho_1 > \rho_2$ with $\rho_1 - \rho_2 \in \mathbb{Z}$, we only get $y_1(x) = \sum_{n\ge0} c_n(x-x_0)^{n+\rho_1}$; the other solution has the form

$$y_2(x) = y_1(x)\ln x + \sum_{n\ge0} b_n x^{n+\rho_2}.$$

Bessel equation & Bessel functions. Consider $x^2y'' + xy' + (x^2-\alpha^2)y = 0$, $\alpha > 0$. Here $p(x) = 1 \Rightarrow p_0 = 1$, $p_j = 0$ $(j \ge 1)$, and $q(x) = x^2 - \alpha^2 \Rightarrow q_0 = -\alpha^2$, $q_2 = 1$, $q_j = 0$ (else). For $y = \sum_{n\ge0} c_n x^{n+\rho}$:

$$f_0(\rho) = \rho(\rho-1) + \rho - \alpha^2 = 0, \qquad f_j(\rho) = p_j\rho + q_j = \begin{cases} 0, & j = 1 \\ 1, & j = 2 \\ 0, & j \ge 3, \end{cases}$$

$$\Rightarrow\; \rho_1 = \alpha,\; \rho_2 = -\alpha,\; \rho_1 - \rho_2 = 2\alpha; \qquad c_{n-2} + n(n+2\rho)c_n = 0.$$

① If $2\alpha \notin \mathbb{Z}$, then $n + 2\rho \neq 0$ for all n: $c_n = -\frac{c_{n-2}}{n(n+2\rho)}$. Taking $c_1 = 0$:

$$c_{2k+1} = 0, \qquad c_{2k} = \frac{(-1)^k c_0}{2^{2k}(\rho+1)(\rho+2)\cdots(\rho+k)\,k!}.$$

With $c_0 = \frac{1}{2^\rho \Gamma(\rho+1)}$ we get the **Bessel function of the first kind**

$$J_\rho(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{\Gamma(k+1+\rho)\,\Gamma(k+1)} \Big(\frac{x}{2}\Big)^{2k+\rho}.$$

Putting $\rho = \pm\alpha$ gives 2 independent solutions.

② If $2\alpha \in \mathbb{Z}$:

2.1 $\alpha = 0$: repeated solutions. With $y_1(x,\rho) = \sum_{n\geq 0} c_n x^{n+\rho} = J_\rho(x)$, another solution is $\frac{\partial}{\partial\rho} y_1(x,\rho)\big|_{\rho=0}$:

$$y_2 = \sum_{n\geq 0} c_n'(\rho) x^{n+\rho} + \sum_{n\geq 0} c_n(\rho) x^{n+\rho} \ln x \,\Big|_{\rho=0} = \sum_{k\geq 1} \frac{(-1)^k}{2^{2k}(k!)^2}(-1)\Big(1 + \frac{1}{2} + \cdots + \frac{1}{k}\Big) x^{2k} + J_0(x)\ln x,$$

$$= J_0(x)\ln x - \sum_{k\geq 1} \frac{(-1)^k}{(k!)^2} \varphi(k) \Big(\frac{x}{2}\Big)^{2k}.$$

2.2 2α odd, i.e. $2\alpha = 2m+1$, $\alpha = m + \frac{1}{2}$: c_{2k} is still well-defined ($2n(2n+2\rho) \neq 0$ for all n); c_{2k+1} is well-defined for $k = 0, \ldots, m-1$, and c_{2m+1} is free (take 0) $\Rightarrow c_{2k+1} = 0$ for all k. We get the same formula as in ①.

2.3 2α even, i.e. $\alpha \in \mathbb{Z}^+$: formally we get the 2nd solution $y_1(x)\ln x + \sum_{n\geq 0} b_n x^{n-\alpha}$.

Rem. (1) **Neumann's function**

$$Y_\alpha(x) = \frac{J_\alpha(x)\cos\alpha\pi - J_{-\alpha}(x)}{\sin\alpha\pi}$$

solves the Bessel equation in every case — the **Bessel function of the second kind**.

(2) $\alpha = \frac{1}{2}$ is a special case.

(3) Asymptotics as $x \to +\infty$:

$$J_\alpha(x) \sim \sqrt{\tfrac{2}{\pi x}}\Big(\cos\big(x - \tfrac{\alpha\pi}{2} - \tfrac{\pi}{4}\big) + o(1)\Big), \qquad Y_\alpha(x) \sim \sqrt{\tfrac{2}{\pi x}}\Big(\sin\big(x - \tfrac{\alpha\pi}{2} - \tfrac{\pi}{4}\big) + o(1)\Big).$$

Chapter 5 Sturm–Liouville theory

We care about **boundary value problems** for second order ODEs.

Motivation. For $u_{tt} - u_{xx} = 0$ with $u'(t,0) = u'(t,L) = 0$, try $u(t,x) = a(t)b(x)$:

$$a''(t)b(x) = a(t)b''(x) \;\Rightarrow\; \frac{a''(t)}{a(t)} = \frac{b''(x)}{b(x)} = -K,$$

$$\left.\begin{cases} a''(t) + Ka(t) = 0 \\ b''(x) + Kb(x) = 0, \quad b'(0) = b'(L) = 0, \end{cases}\right\} \text{a boundary value problem.}$$

Taking $\mathcal{L} := \frac{d^2}{dx^2}$, consider $\mathcal{L}b = -Kb$, $b'(0) = b'(L) = 0$. From $b'' + Kb = 0$: if $K = 0$, $b = c_1 x + c_2$ and $b'(0) = b'(L) = 0$ gives $c_1 = 0$; if $K < 0$, $b = c_1 e^{\sqrt{-K}x} + c_2 e^{-\sqrt{-K}x}$ and the boundary conditions force $c_1 = c_2 = 0$ (trivial); if $K > 0$, $b = c_1 \cos\sqrt{K}x + c_2 \sin\sqrt{K}x$ with $b'(0) = c_2\sqrt{K} = 0 \Rightarrow c_2 = 0$ and $b'(L) = c_1\sqrt{K}\sin\sqrt{K}L$: if $\sqrt{K}L = n\pi$, $c_1 \neq 0$ works, so

$$K = \left(\frac{n\pi}{L}\right)^2.$$

For this problem there are infinitely many $K_n \nearrow +\infty$ with eigenfunctions $\varphi(x, K_n)$; each $\varphi(x, K_n)$ has exactly n zeros in $(0, L)$.

5.1 Sturm separation & comparison thm.

Consider $y'' + p(x)y' + q(x)y = 0$ on an interval $J \subseteq \mathbb{R}$, $p, q \in C(J)$. (Margin: $y'' + py' + qy = f(x)$ has general solution $\varphi_1(x)\int \frac{W_1(x)}{W(x)} f(x) + \varphi_2(x)\int \frac{W_2(x)}{W(x)} f(x)$, with $\begin{vmatrix} 0 & 1 \\ -p & -q \end{vmatrix}$.)

Thm ((Separation thm))**.** Consider $y'' + p(x)y' + q(x)y = 0$ on J.

(1) For any nonzero solution $\varphi(x)$, every zero of $\varphi(x)$ is **isolated**.

(2) For any two nonzero solutions φ_1, φ_2: ① φ_1, φ_2 dependent $\iff$ they have the same zeros; ② φ_1, φ_2 independent $\iff$ they have alternate zeros (interlace strictly).

(Hint: construct $\varphi_1'\varphi_2 - \varphi_1\varphi_2'$.)

Pf. (1) If zeros $x_j \to x_0 \in J$, then $\varphi'(x_0) = \lim_{j\to\infty} \frac{\varphi(x_j) - \varphi(x_0)}{x_j - x_0} = 0$; with $\varphi(x_0) = 0$ this gives $\varphi \equiv 0$. □

(2) ① ($\Rightarrow$) $\varphi_2 = c\varphi_1$ ($c \neq 0$): same zeros. ($\Leftarrow$) Consider $W(\varphi_1, \varphi_2) = \varphi_1\varphi_2' - \varphi_1'\varphi_2$. At a common zero x_0: $W(\varphi_1, \varphi_2)(x_0) = 0$; since $W(x) = W(x_0)e^{\int_{x_0}^x \operatorname{tr} A(s)ds}$ with $\operatorname{tr} A(s) = -q$ (for $\begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -p & -q \end{pmatrix}\begin{pmatrix} y \\ y' \end{pmatrix}$), $W \equiv 0$; hence dependent. □

② ($\Leftarrow$) by ①. ($\Rightarrow$) Claim: for any 2 consecutive zeros $x_1 < x_2$ of $\varphi(x)$ there is $c \in (x_1, x_2)$ with $\varphi_2(c) = 0$. Suppose $\varphi_2 \neq 0$ on (x_1, x_2) (it is $\neq 0$ on $[x_1, x_2]$, being independent). Consider $\psi(x) = \frac{\varphi_1(x)}{\varphi_2(x)} \in C'[x_1, x_2]$:

$$\psi(x_1) = \psi(x_2) = 0 \;\Rightarrow\; \exists c \in (x_1, x_2) : \psi'(c) = 0 \;\Rightarrow\; \varphi_1'(c)\varphi_2(c) - \varphi_1(c)\varphi_2'(c) = 0 \;\Rightarrow\; W(c) = 0,$$

so φ_1, φ_2 dependent: contradiction. □

Thm ((Sturm comparison))**.** Let

$$y'' + P(x)y' + Q_1(x)y = 0 \ (*) \qquad \text{and} \qquad y'' + P(x)y' + Q_2(x)y = 0 \ (**)$$

on J with $P, Q_1, Q_2 \in C(J)$ ($Q_1 \leq Q_2 \Rightarrow Q_2$ waves faster $\Rightarrow$ more zeros). Suppose the nonzero $\varphi(x)$ solves $(*)$ and has consecutive zeros $x_1 < x_2$. If $Q_1(x) \leq Q_2(x)$, then every solution $\psi(x)$ of $(**)$ has a zero $c \in [x_1, x_2]$ (if $Q_1 < Q_2$, then $c \in (x_1, x_2)$).

Pf. Suppose $\varphi > 0$ on (x_1, x_2); then $\varphi'(x_1) > 0$, $\varphi'(x_2) < 0$. Let ψ solve $(**)$ and suppose $\psi \neq 0$ on $[x_1, x_2]$; WLOG $\psi > 0$ on (x_1, x_2). From $\psi(\text{I}) - \varphi(\text{II})$ where (I), (II) are the two equations: $v(x) = \psi'\varphi - \psi\varphi'$ satisfies

$$v'(x) + P(x)v(x) = (Q_2 - Q_1)\varphi\psi \;\Rightarrow\; \left(v(x)e^{\int_{x_1}^x P(s)ds}\right)' = (Q_2 - Q_1)\varphi\psi\, e^{\int_{x_1}^x P(s)ds} \geq 0,$$

so $v(x_2)e^{\int_{x_1}^{x_2} P} \geq v(x_1)$. But $v(x_2) = \varphi'(x_2)\psi(x_2) < 0$ and $v(x_1) = \varphi'(x_1)\psi(x_1) > 0$: contradiction. So ψ has a zero in $[x_1, x_2]$. □

Def. Consider $\mathcal{L} := \frac{d^2}{dx^2} + p(x)\frac{d}{dx} + q(x)$ (or $A(x)\frac{d^2}{dx^2} + B(x)\frac{d}{dx} + C(x)$). If a nontrivial solution φ of $\mathcal{L}\varphi = 0$ on J has infinitely many zeros in J, we call φ an **oscillating solution** on J.

Rem. If one φ oscillates, every nontrivial solution oscillates: for any nontrivial ψ, either $\psi = c\varphi$ or ψ, φ have alternate zeros. So we call $\mathcal{L}$ oscillating.

Ex. If $q(x) \leq 0$ on J, any solution of $y'' + p(x)y' + q(x)y = 0$ has at most 1 zero on J.

Pf. Compare (I) $y'' + py' + qy = 0$ with (II) $y'' + p(x)y' = 0$: if (I) had 2 zeros $x_1 < x_2$, any solution of (II) would have a zero in $[x_1, x_2]$; but $y \equiv 1$ solves (II). Contradiction. □

Ex. $y'' + Q(x)y = 0$ with $Q \in C[a, +\infty)$ & $Q(x) \geq m > 0$: any solution is oscillating, and the gap between consecutive zeros is $< \frac{\pi}{\sqrt{m}}$.

Pf. Compare with $y'' + my = 0$, which has the solution $y = \cos\sqrt{m}x$ with infinite zeros & gap $= \frac{\pi}{\sqrt{m}}$. □

Ex. $x^2y'' + xy' + (x^2 - \alpha^2)y = 0$, $\alpha \geq 0$: the solutions are oscillating. Rewrite: $y'' + \frac{1}{x}y' + (1 - \frac{\alpha^2}{x^2})y = 0$. The Euler equation $y'' + \frac{1}{x}y' + \frac{c}{x^2}y = 0$ has (trying x^λ) $\lambda = \pm i\sqrt{c}$, i.e. fundamental solutions $\cos(\sqrt{c}\ln x)$, $\sin(\sqrt{c}\ln x)$. We want $1 - \frac{\alpha^2}{x^2} > \frac{c}{x^2}$, i.e. choose N s.t. this holds on $[N, +\infty)$. Then by comparison the original solution is oscillating. □

Ex. $y'' + p(x)y = 0$ with $p \in C(\mathbb{R})$ and $n^2 < p(x) < (n+1)^2$: no nontrivial solution is 2π-periodic.

Pf. Suppose φ is 2π-periodic; subtracting the equations at x and $x + 2\pi$ gives $\varphi(x)[p(x + 2\pi) - p(x)] = 0$. The zero set I_φ has only isolated points, so $p(x) = p(x + 2\pi)$ off I_φ; since I_φ^C is dense and $p \in C(\mathbb{R})$, $p(x) = p(x + 2\pi)$ everywhere. So p is 2π-periodic and attains its max & min: $\exists\delta$ with $(n+\delta)^2 < p(x) < (n+1-\delta)^2$. Comparing with $y'' + (n+\delta)^2 y = 0$ and $y'' + (n+1-\delta)^2 y = 0$, consecutive zeros of φ have gaps in $\left[\frac{\pi}{n+1-\delta}, \frac{\pi}{n+\delta}\right]$. If x^* is a zero, so is $x^* + 2\pi$, and the number of gaps is even (to verify $\varphi'(x^*) = \varphi'(x^* + 2\pi)$). Hence

$$2\pi = \sum_{i=1}^{2k}(x_i - x_{i-1}) \in \left[\frac{k\pi}{n+1-\delta}, \frac{k\pi}{n+\delta}\right] \quad \Rightarrow \quad n + \delta \leq k \leq n + 1 - \delta,$$

a contradiction. □

5.2 Sturm & Liouville problem

Let $\mathcal{L} := A(x)\frac{d^2}{dx^2} + B(x)\frac{d}{dx} + C(x)$ with $A(x) > 0$ on $J = [a, b]$. We want to solve the **eigenvalue problem** $\mathcal{L}y = \lambda y$ $(*)$ together with the **boundary value problem**

$$\begin{cases} Ry(a) + Ly'(a) = 0 \\ My(b) + Ny'(b) = 0. \end{cases}$$

Def. A condition like $y(a) = 0$ is called a **Dirichlet problem**; $y'(a) = 0$ a **Neumann problem**; the general case $Ky(a) + Ly'(b) = 0$ a **Robin problem**.

Reduction 1. There are $p(x) \in C'$, $q(x), r(x) \in C$ with $p(x), r(x) > 0$ s.t. $(*)$ becomes

$$\mathcal{L}y = \frac{1}{r(x)}\big((p(x)y')' + q(x)y\big) = \lambda y(x),$$

namely $p(x) = \exp\big(\int_{x_0}^{x} \frac{B(s)}{A(s)} ds\big)$, $r(x) = p(x)/A(x)$, $q(x) = r(x)C(x)$.

Reduction 2. Change variables $x = x(t)$ with $x(0) = a$, $x(1) = b$, choosing $\frac{dx}{dt} = c_0 p(x)$ where c_0 is solved from $1 = \int_a^b \frac{ds}{c_0 p(s)}$. Then $y_t'' = (c_0 p(x) y'(x))' \, c_0 p(x)$ and

$$\mathcal{L}y = \frac{1}{r(x)}\Big(\frac{d^2 y}{dt^2} \cdot \frac{1}{c_0^2 p(x)} + q(x)y\Big).$$

So consider $\widetilde{\mathcal{L}} := \frac{1}{\widetilde{r}(t)}\Big(\frac{d^2}{dt^2} + \widetilde{q}(t)\Big)$ with $\widetilde{r}(t) = c_0^2 p(x(t)) r(x(t)) > 0$, $\widetilde{q}(t) = c_0^2 p(x(t)) q(x(t))$. The eigenvalue problem becomes $\widetilde{\mathcal{L}}y = -\lambda y \iff y''(t) + (\lambda\, r(t) + q(x(t)))y(t) = 0$, and the boundary conditions become

$$\cos\alpha \cdot y(0) - \sin\alpha \cdot y'(0) = 0, \qquad \cos\beta \cdot y(1) - \sin\beta \cdot y'(1) = 0$$

(assume $0 \le \alpha < \pi$, $0 \le \beta < \pi$).

Finally, the **Sturm–Liouville Problem**:

$$\begin{cases} y'' + (\lambda\, r(x) + q(x))\, y(x) = 0 & \text{on } [0,1], \\ \cos\alpha \cdot y(0) - \sin\alpha \cdot y'(0) = 0, \\ \cos\beta \cdot y(1) - \sin\beta \cdot y'(1) = 0. \end{cases}$$

Functional analytic setup. Let $\mathcal{F}$ be $\mathbb{R}$ or $\mathbb{C}$, X a vector space over $\mathcal{F}$, and $\langle\cdot,\cdot\rangle : X \times X \to \mathcal{F}$ satisfying ① $\langle a, b\rangle = \langle b, a\rangle$ on $\mathbb{R}$ (or $\langle a, b\rangle = \overline{\langle b, a\rangle}$ on $\mathbb{C}$); ② linearity in the first argument; ③ $\langle a, a\rangle \ge 0$ with equality only at $a = 0$. Then $(X, \langle\cdot,\cdot\rangle)$ is an **inner product space**; the distance is $d(f, g) := \langle f - g, f - g\rangle$; if every Cauchy sequence converges in X, X is complete — a complete inner product space is a **Hilbert space**. $T : X \supseteq D(T) \to X$ is **linear** if $T(\lambda a + \mu b) = \lambda T(a) + \mu T(b)$, and **symmetric** if $\langle Ta, b\rangle = \langle a, Tb\rangle$.

Eg. $X = \{f \in C[0,1]\}$, $\langle f, g\rangle = \int_0^1 fg\, dx$; $T = \frac{d^2}{dx^2}$ with $D(T) = \{f \in C^2[0,1] : f'(0) = f(0) = 1\}$:

$$\langle Tf, g\rangle = \int_0^1 f'' g\, dx = (f'g - fg')\big|_0^1 + \int_0^1 fg''\, dx = \langle f, Tg\rangle,$$

so T is symmetric. With the domain $D_1 = \{f \in C^2[0,1] : \cos\alpha\, f(0) - \sin\alpha\, f'(0) = 0,\ \cos\beta\, f(1) - \sin\beta\, f'(1) = 0\}$, T on D_1 is also symmetric.

For the Sturm–Liouville problem take $X := \{f \in C[0,1]\}$, $\mathcal{F} = \mathbb{R}$, $\langle f, g\rangle := \int_0^1 r(x) f(x) g(x)\, dx$, and $T := \frac{1}{r(x)}\big(\frac{d^2}{dx^2} + q(x)\big)$ with domain the $f \in C^2[0,1]$ satisfying the boundary conditions. Then T is symmetric.

Thm. For the Sturm–Liouville problem $y'' + (\lambda r + q)y = 0$ with the boundary conditions:

(1) There exist countably many eigenvalues $\lambda_1 < \lambda_2 < \cdots \to +\infty$.

(2) Each λ_n has a 1-dim eigenspace; the eigenfunction $\varphi(x, \lambda_n)$ has exactly n zeros in $[0,1]$.

(3) The $\varphi(x, \lambda_n)$ are orthogonal: $\int_0^1 r(x)\varphi(x,\lambda_n)\varphi(x,\lambda_m)dx = 0$ for $n \ne m$ ($= \delta_n^2$ for $n = m$), and complete: $\int_0^1 r(x) f(x) \varphi(x,\lambda_n) dx = 0\ \forall n \Rightarrow f \equiv 0$; moreover $\forall f \in X$: $f = \sum_{n=0}^{\infty} c_n \varphi(x, \lambda_n)$, uniformly on compact intervals if $f \in D(\mathcal{L})$.

(4) Between any consecutive zeros of φ_n there is a zero of φ_m (for $\lambda_m > \lambda_n$).

Pf. Consider the solution φ with $\varphi(0) = \sin\alpha$, $\varphi'(0) = \cos\alpha$ $(*)$; by Picard there is a unique $\varphi(x,\lambda)$ solving the ODE and the 1st boundary condition. The goal is to find λ_0 s.t. $\varphi(x,\lambda_0)$ satisfies the 2nd boundary condition. Use the **Prüfer transform**:

$$\varphi(x,\lambda) = \rho(x,\lambda)\sin\theta(x,\lambda), \qquad \varphi'(x,\lambda) = \rho(x,\lambda)\cos\theta(x,\lambda), \qquad \theta(0,\lambda) = \alpha \in [0,\pi).$$

Then $\theta(x,\lambda) = \arctan\frac{\varphi}{\varphi'}$ is a well-defined continuous function with

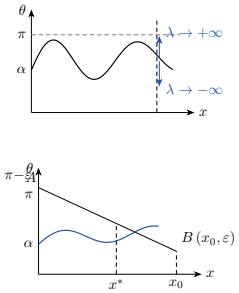
$$\theta' = \cos^2\theta + (\lambda r+q)\sin^2\theta, \qquad \begin{cases}\theta' = \cos^2\theta + (\lambda r+q)\sin^2\theta\\ \theta(0,\lambda) = \alpha\end{cases} \quad \& \quad \begin{cases}\rho' = \frac{1}{2}\rho\,(1-\lambda r-q)\sin 2\theta\\ \rho(0,\lambda) = 1.\end{cases}$$

$\varphi(x,\lambda)$ satisfies the 2nd boundary condition $\iff$ $\theta(1,\lambda) = \beta + k\pi$.

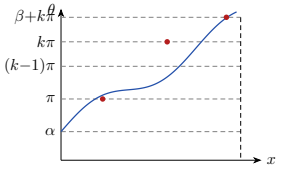
Lemma 1. For $x \in (0,1]$, $\theta(x,\lambda)$ is C^1 & strictly increasing in λ. (Pf. from C^1 dependence: $\frac{d}{dx}\frac{\partial\theta}{\partial\lambda} = (\lambda r+q-1)\sin\theta\,\frac{\partial\theta}{\partial\lambda} + r\sin^2\theta$, $\frac{\partial\theta}{\partial\lambda}(0,\lambda) = 0$, giving $\frac{\partial\theta}{\partial\lambda} > 0$.) #

Lemma 2. For $x_0 \in (0,1]$: $\theta(x_0,\lambda) > 0$ for every λ. (Pf. $\theta(0,\lambda) = \alpha$, $\theta'(0,\lambda) = 1$ if $\alpha = 0$; find x_1 with $\theta > 0$ on $(0,x_1]$; if $x_1 \neq 1$, at the first zero x^* we'd have $\theta'(x^*,\lambda) \le 0$ but the equation gives $\theta'(x^*,\lambda) = 1$: contradiction.) #

Lemma 3. For $x_0 \in (0,1]$: $\theta(x_0,\lambda) \to 0$ as $\lambda \to -\infty$ and $\theta(x_0,\lambda) \to +\infty$ as $\lambda \to +\infty$. (Pf. $\theta' = \cos^2\theta + (\lambda r+q)\sin^2\theta$; for $\lambda < 0$, $\lambda r+q < \lambda m+M$. Given ε with $\alpha < \pi-\varepsilon$, let $A = (0,\pi-\varepsilon)$, $B = (x_0,\varepsilon)$; for λ very negative the solution stays below the line AB — otherwise at the first intersection x^*, $\theta'(x^*,\lambda) \ge k_{AB} = \frac{2\varepsilon-\pi}{x_0}$ while $\theta(x^*,\lambda) \in (\varepsilon,\pi-\varepsilon)$ gives $\theta' \le 1 + (\lambda m+M)\sin^2\varepsilon$, a contradiction as $\lambda \to -\infty$. Hence $\theta(x_0,\lambda) \to 0$; similarly $\theta(x_0,\lambda) \to +\infty$.) #

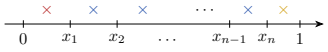


Now for each $k \in \mathbb{N}$ there is λ_k s.t. $\theta(1,\lambda_k) = \beta + k\pi$; the $\lambda_k \nearrow +\infty$ and each λ_k is unique. This gives all eigenvalues λ_k & eigenfunctions $\varphi(x,\lambda_k)$. Every eigenspace is 1-dim because the initial data are linearly dependent $\Rightarrow$ solutions are dependent.



Zeros. $\varphi(x^*,\lambda_k) = 0 \iff \sin\theta(x^*,\lambda_k) = 0$. For $k \ge 1$ there are k zeros $x_j \in (0,1)$; at a zero, $\theta'(x_j,\lambda_k) = \cos\theta + (\lambda_k r+q)\sin\theta \equiv 1$, so each level $j\pi$ is crossed only once. For $k = 0$, $\theta(x,\lambda_0) = j\pi$ has no solution (checking the sketches for $\beta = \pi$ and $\beta < \pi$).

Oscillation of zeros. For $\lambda_m > \lambda_n$ we have $\lambda_m r+q > \lambda_n r+q$; by Sturm comparison, between 2 consecutive zeros of $\varphi(x,\lambda_n)$ there is a zero of $\varphi(x,\lambda_m)$.



For $\varphi(x,\lambda_n)$ & $\varphi(x,\lambda_{n+1})$, let $0 < x_1 < \cdots < x_n < 1$ be the zeros of $\varphi(x,\lambda_n)$. In any (x_k,x_{k+1}) there is a zero of $\varphi(x,\lambda_{n+1})$. In $(0,x_1)$: $\theta(x_1,\lambda_{n+1}) > \theta(x_1,\lambda_n) = \pi$ while $\theta(0,\lambda_{n+1}) = \alpha < \pi$, so there is $x^* \in (0,x_1)$ with $\theta(x^*,\lambda_{n+1}) = \pi$. In $(x_n,1)$: if $\varphi(x,\lambda_{n+1}) \ge 0$ on $[x_n,1]$, the proof of Sturm comparison gives a contradiction; so $\exists\, x^{**} \in (x_n,1)$ with $\varphi(x^{**},\lambda_{n+1}) = 0$. For orthogonality, notice the λ_k are different. □

5.3 Periodic Sturm–Liouville problem

Consider $y'' + (\lambda r + q)y = 0$, $r, q \in C[0,1]$, $r > 0$, with r, q 1-periodic; ask for 1-periodic solutions.

General setting: $y'' = f(x, y, y')$ with f locally Lipschitz in y, y' and 1-periodic in x.

Lemma. Any solution satisfying $y(0) = y(1)$, $y'(0) = y'(1)$ is a 1-periodic solution.

Pf. $y(x+1)$ satisfies the same IVP as $y(x)$. □

Consider $\mathcal{L} := \frac{1}{r(x)}\big(\frac{d^2}{dx^2} + q(x)\big)$, $r > 0$, $r, q \in C[0,1]$ 1-periodic; $X := \{f \in C^2[0,1]\}$, $\langle f, g\rangle = \int_0^1 r(x)f(x)g(x)\,dx$,

$$D(\mathcal{L}) := \{f \in X :\ f(0) = f(1)\ \&\ f'(0) = f'(1)\}.$$

$\mathcal{L}$ is symmetric on $D(\mathcal{L})$; so we study $\mathcal{L}y = -\lambda y$.

Special case. $r \equiv 1$, $q \equiv 0$.

Thm ((Periodic Sturm–Liouville problem)). (1) The eigenvalues are countable: $\lambda_0 < \lambda_1 \le \lambda_2 < \lambda_3 \le \lambda_4 < \cdots < \lambda_{2k-1} \le \lambda_{2k} < \cdots$.

(2) λ_0 has a 1-dim eigenspace.

(3) If $\lambda_{2k-1} < \lambda_{2k}$, each of them has a 1-dim eigenspace.

(4) If $\lambda_{2k-1} = \lambda_{2k} = \lambda$, then λ has a 2-dim eigenspace.

(5) $\varphi(x, \lambda_0)$ has no zero in $[0,1)$; $\varphi(x, \lambda_{2k-1})$ & $\varphi(x, \lambda_{2k})$ have $2k$ zeros in $[0,1)$.

(6) Orthogonality & completeness hold.

Pf. Step 1 (Foundation): reconstruct the boundary conditions. Take fundamental solutions $\begin{pmatrix}\varphi(0)\\ \varphi'(0)\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$, $\begin{pmatrix}\psi(0)\\ \psi'(0)\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}$; any solution is $y = c_1\varphi + c_2\psi$. The boundary conditions give

$$\begin{pmatrix}1-\varphi(1,\lambda) & -\psi(1,\lambda)\\ -\varphi'(1,\lambda) & 1-\psi'(1,\lambda)\end{pmatrix}\begin{pmatrix}c_1\\c_2\end{pmatrix} = 0, \qquad \det A(\lambda) = 2 - \varphi(1,\lambda) - \psi'(1,\lambda)$$

(using $W(\varphi,\psi) = 1$). Denote $f(\lambda) = \varphi(1,\lambda) + \psi'(1,\lambda)$: λ is an eigenvalue $\iff$ $f(\lambda) = 2$. Moreover, λ gives 2 independent solutions $\iff$ $A(\lambda) \equiv 0$ $\iff$ $\varphi(1,\lambda) = 1$, $\psi(1,\lambda) = 0$, $\varphi'(1,\lambda) = 0$, $\psi'(1,\lambda) = 1$.

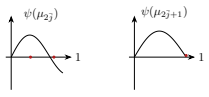
Step 2 (Preparation). For (I) $y'' + (\lambda r + q)y = 0$, $y(0) = y(1) = 0$, the S–L thm gives $\mu_0 < \mu_1 < \cdots \to +\infty$ with $\psi(x, \mu_j)$ solving (I). For (II) $y'' + (\lambda r + q)y = 0$, $y'(0) = y'(1) = 0$, it gives $\nu_0 < \nu_1 < \cdots \to +\infty$ with $\varphi(x, \nu_j)$ solving (II).

Step 3 (Properties of $f(\lambda)$).

Lemma. $\nu_0 \le \mu_0$, $f(\nu_0) \ge 2$, $f(\mu_{2j+1}) \ge 2$, $f(\mu_{2j}) \le -2$.

Pf. $1 = W(\varphi,\psi)(1) = \varphi(1,\lambda)\psi'(1,\lambda) - \varphi'(1,\lambda)\psi(1,\lambda)$. For $\lambda = \nu_0$: $\varphi'(1,\nu_0) = 0 \Rightarrow \varphi(1,\nu_0)\psi'(1,\nu_0) = 1$. For $\lambda = \mu_j$: $\psi(1,\mu_j) = 0 \Rightarrow \varphi(1,\mu_j)\psi'(1,\mu_j) = 1$, so

$$f(\mu_j) = \psi'(1,\mu_j) + \frac{1}{\psi'(1,\mu_j)}.$$



Counting zeros of $\psi(\cdot, \mu_j)$ on $[0,1]$: $\psi'(1,\mu_{2j}) < 0 \Rightarrow f(\mu_{2j}) \le -2$; $\psi'(1,\mu_{2j+1}) > 0 \Rightarrow f(\mu_{2j+1}) \ge 2$. For $\lambda = \nu_0$: $f(\nu_0) = \varphi(1,\nu_0) + \frac{1}{\varphi(1,\nu_0)}$ with $\varphi(1,\nu_0) > 0$, so $f(\nu_0) \ge 2$. If $\nu_0 > \mu_0$: $\psi(x,\mu_0)$ has 2 zeros in $[0,1]$, so by comparison $\varphi(x,\nu_0)$ has 1 zero: contradiction. □

Step 4 (Properties of $f'(\lambda)$). Differentiating the ODEs in λ: with $u = \frac{\partial \varphi}{\partial \lambda}$, $u'' + (\lambda r + q)u = -r\varphi$, and using $u(0) = u'(0) = 0$,

$$\frac{\partial}{\partial \lambda}\varphi(1,\lambda) = \int_0^1 \big(\varphi(1)\psi(s) - \psi(1)\varphi(s)\big) r(s)\varphi(s)\, ds, \qquad \frac{\partial}{\partial \lambda}\psi'(1,\lambda) = \int_0^1 \big(\varphi'(1)\psi(s) - \psi'(1)\varphi(s)\big) r(s)\psi(s)\, ds.$$

So

$$f'(\lambda) = \int_0^1 \big[a\,\psi^2(s,\lambda) + b\,\psi(s,\lambda)\varphi(s,\lambda) + c\,\varphi^2(s,\lambda)\big] r(s)\, ds,$$

with $a := \varphi'(1,\lambda)$, $b := \varphi(1,\lambda) - \psi'(1,\lambda)$, $c := -\psi(1,\lambda)$. Notice

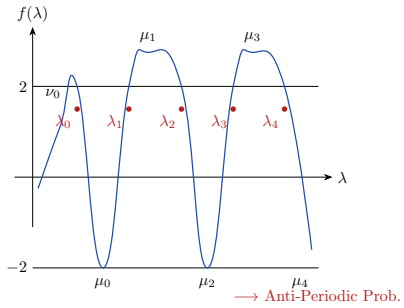
$$b^2 - 4ac = \varphi^2(1) + \psi'^2(1) - 2\varphi(1)\psi'(1) + 4\varphi'(1)\psi(1) = \varphi^2(1) + \psi'^2(1) + 2\varphi(1)\psi'(1) - 4 = (f(\lambda))^2 - 4.$$

If $f(\lambda) = 2$, then $b^2 - 4ac = 0$; and $-\psi(1,\lambda) \geq 0 \Rightarrow f'(\lambda) \geq 0$, $-\psi(1,\lambda) \leq 0 \Rightarrow f'(\lambda) \leq 0$, with $f'(\lambda) = 0 \iff a, b, c = 0 \iff A(\lambda) = 0$. Hence, for an eigenvalue,

$$f'(\lambda) = 0 \iff \text{the eigenspace of } \lambda \text{ is 2-dim.}$$

Also, for eigenvalues $\lambda \neq \mu_j$: $\lambda < \mu_0 \Rightarrow \psi(1,\lambda) > 0 \Rightarrow f'(\lambda) < 0$; $\mu_0 < \lambda < \mu_1 \Rightarrow f'(\lambda) > 0$; and $\lambda \in (\mu_j, \mu_{j+1}) \Rightarrow f'(\lambda)(-1)^j < 0$. Besides, one computes: if $f(\mu_{2j+1}) = 2$ and $f'(\mu_{2j+1}) = 0$, then $f''(\mu_{2j+1}) < 0$.

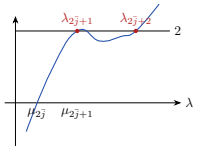
Step 5 (Existence of eigenvalues). We have $\nu_0 \leq \lambda_0 < \mu_0 < \lambda_1 \leq \mu_1 \leq \lambda_2 < \mu_2 < \lambda_3 \leq \mu_3 \leq \lambda_4$:



Case 1. If $f(\mu_{2j+1}) > 2$, then $\lambda_{2j+1} < \mu_{2j+1} < \lambda_{2j+2} < \mu_{2j+2}$, and λ is unique in (μ_{2j+1}, μ_{2j+2}) since $f' \neq 0$.

Case 2. If $f(\mu_{2j+1}) = 2$:

Case 2.1 $f'(\mu_{2j+1}) \neq 0$ (say > 0): take $\lambda_{2j+1} = \mu_{2j+1}$ and find $\lambda_{2j+2} \in (\mu_{2j+1}, \mu_{2j+2})$, unique with $f'(\lambda_{2j+2}) \neq 0$; the eigenspace is 1-dim.



Case 2.2 $f'(\mu_{2j+1}) = 0$: $\lambda = \mu_{2j+1}$ has a 2-dim eigenspace; take $\lambda_{2j+1} = \mu_{2j+1} = \lambda_{2j+2}$, and no other λ in $(\mu_{2j}, \mu_{2j+1}) \cup (\mu_{2j+1}, \mu_{2j+2})$ satisfies $f(\lambda) = 2$.

Conclusion: $\lambda_{2j+1} < \lambda_{2j+2} \Rightarrow$ 1-dim eigenspace; $\lambda_{2j+1} = \lambda_{2j+2} \Rightarrow$ 2-dim eigenspace.

Step 6 (Number of zeros).

Lemma. If λ is an eigenvalue, $\varphi(x,\lambda)$ has an even number of zeros in $[0,1)$.

Pf. Zeros are isolated, hence finite. If $\varphi(0,\lambda) = 0$, then $\varphi(1,\lambda) = 0$ and $\varphi'(0,\lambda) = \varphi'(1,\lambda)$; counting gives an even number. If $\varphi(0,\lambda) \neq 0$, same argument. □

Now $\varphi(x,\lambda_0)$ has no zero in $[0,1)$: otherwise it would have at least 2 zeros, but $\lambda_0 < \mu_0$ and comparison give a contradiction. And $\varphi(x,\lambda_{2j+1})$, $\varphi(x,\lambda_{2j+2})$ both have $2j+2$ zeros: by comparison with $\mu_{2j} < \lambda_{2j+1} \leq \mu_{2j+1} \leq \lambda_{2j+2} < \mu_{2j+2}$ (where $\psi(x,\mu_{2j})$ has $2j$ zeros, $\psi(x,\mu_{2j+1})$ has $2j+1$, $\psi(x,\mu_{2j+2})$ has $2j+2$), they have between $2j+1$ and $2j+3$ zeros; being even, both have exactly $2j+2$ zeros in $[0,1)$. □

5.4 BVP for nonlinear ODEs

Consider $\begin{cases} y'' = f(x, y, y') \\ y(0) = a,\ y(1) = b \end{cases}$ with f Lipschitz in y & y'.

5.4.1 Shooting method

Solve $\begin{cases} y'' = f(x, y, y') \\ y(0) = a,\ y'(0) = s \end{cases}$ $\Rightarrow \varphi(x, s)$; we want $\varphi(1, s) = b$. Consider $F(s) = \varphi(1, s) - b$ (s is a choice).

Thm. If the IVP can be solved on $[0, 1]$ and $\exists\ s_1 > s_2$ with $F(s_1)F(s_2) < 0$, then $\exists\ s^*$ s.t. $\varphi(x, s^*)$ solves the BVP.

Eg. $\begin{cases} x''(t) = \arctan x(t), \quad t \in [0, 1] \\ x(0) = 0,\ x(1) = 1. \end{cases}$

For the IVP $x(0) = 0$, $x'(0) = \lambda$ the solution $\varphi(x, \lambda)$ exists on $[0, 1]$ by the extension thm. Let $F(\lambda) := \varphi(1, \lambda) - 1$. For $\lambda = 0$: $x(t) \equiv 0$, so $F(0) = -1 < 0$. For $\lambda = 1$: let $T := \sup\{t_0 \in [0, 1] : x(t) \geq 0 \text{ on } [0, t_0]\}$; then $x(t) \geq 0$ on $[0, T] \Rightarrow x'' \geq 0 \Rightarrow x'(t) \geq x'(0) = 1$, so

$$x(t) - x(0) = \int_0^t x'(t)\, dt \geq t \quad \Rightarrow \quad x(t) \geq t.$$

In particular $x(T) \geq T > 0$, hence $T = 1$; so $x(1) \geq 1$, i.e. $F(1) \geq 0$. There is λ^* with $F(\lambda^*) = 0$, giving a solution of the BVP. □

5.4.2 Periodic BVP

Consider $\begin{cases} y'' = f(x, y, y') \\ y(0) = y(1),\ y'(0) = y'(1) \end{cases}$ with $f(x + 1, \cdot, \cdot) = f(x, \cdot, \cdot)$.

The IVP with $y(0) = \alpha$, $y'(0) = \beta$ gives $\varphi(1, \alpha, \beta)$; we want $\varphi(1, \alpha, \beta) = \alpha$, $\varphi'(1, \alpha, \beta) = \beta$. The **Poincaré map** $P(\alpha, \beta) = \big(\varphi(1, \alpha, \beta),\ \varphi'(1, \alpha, \beta)\big)^T$ should have a **fixed point**.

Methods for finding fixed points: (1) Banach's fixed point thm.; (2) Brouwer's fixed point thm. (continuous $P : D \to D$ with D compact & convex); (3) the implicit function thm (locally).

Eg. $x'' + \omega_0^2 x = p(t) + \varepsilon f(t, x, x')$ with $\omega_0 > 0$, $\frac{\omega_0}{2\pi} \notin \mathbb{Z}$, p 1-periodic, $f(t + 1, \cdot, \cdot) = f(t, \cdot, \cdot)$. If ε is small enough, the equation has a 1-periodic solution.

Pf. For the IVP $x(0) = x_0$, $x'(0) = v_0$ we have the unique solution $\varphi(x; x_0, v_0, \varepsilon)$; we want

$$\begin{cases} \varphi(1, x_0, v_0, \varepsilon) = x_0 \\ \varphi'(1, x_0, v_0, \varepsilon) = v_0 \end{cases} \iff \begin{cases} F := \varphi(1, x_0, v_0, \varepsilon) - x_0 = 0 \\ G := \varphi'(1, x_0, v_0, \varepsilon) - v_0 = 0. \end{cases} \qquad (*)$$

For $\varepsilon = 0$:

$$x(t, x_0, v_0, 0) = x_0 \cos \omega_0 t + \frac{v_0}{\omega_0} \sin \omega_0 t + \frac{1}{\omega_0} \int_0^t p(s) \sin \omega_0 (t - s)\, ds.$$

Substituting into $(*)$, the determinant $\Delta \neq 0$; so $(*)$ is solved by some (x_0^*, v_0^*). Since $\frac{\partial(F,G)}{\partial(x_0,v_0)}\big|_{(x_0^*, v_0^*)} = \Delta \neq 0$, a vector function $(x_0(\varepsilon), v_0(\varepsilon))$ exists. By the implicit function thm, in a small neighborhood of (x_0^*, v_0^*) & $\varepsilon = 0$ there are $x_0(\varepsilon), v_0(\varepsilon)$ with $F \equiv G \equiv 0$, giving a 1-periodic solution for all small ε. □

Eg. $x'' + x + C \arctan x = \sin t$ with $|C| > \frac{1}{2}$ has a 2π-periodic solution.

Pf. Step 1 (set up). The homogeneous equation $x'' + x = 0$ has solutions $\sin t, \cos t$ with $W(t) = \begin{vmatrix} \sin t & \cos t \\ \cos t & -\sin t \end{vmatrix} = -1$. Consider the solution $\varphi_b(t)$ with $\varphi_b(0) = 0$, $\varphi_b'(0) = b$. Since $-\varphi_b(t)$ satisfies the same IVP, $\varphi_b(t) \equiv -\varphi_b(-t)$; so consider $f(b) = \varphi_b(\pi) - \varphi_b(-\pi)$.

Step 2 (properties of $f(b)$). By variation of parameters:

$$\varphi_b(t) = b\sin t + \int_0^t \sin(t-s)\big[\sin s - C\arctan x(s)\big]\,ds,$$

so

$$f(b) = \int_{-\pi}^{\pi} \sin^2 s\,ds - C\int_{-\pi}^{\pi} \sin s\,\arctan x(s)\,ds = \pi - C\int_{-\pi}^{\pi} \sin s\,\arctan x(s)\,ds.$$

Splitting the last integral,

$$\int_{-\pi}^{\pi} * = \int_{-\pi}^{-\pi+\varepsilon} + \int_{-\pi+\varepsilon}^{-\varepsilon} + \int_{-\varepsilon}^{\varepsilon} + \int_{\varepsilon}^{\pi-\varepsilon} + \int_{\pi-\varepsilon}^{\pi} = \int_{-\pi+\varepsilon}^{-\varepsilon} + \int_{\varepsilon}^{\pi-\varepsilon} + O(\varepsilon),$$

$$\int_{\varepsilon}^{\pi-\varepsilon} * = \int_{\varepsilon}^{\pi-\varepsilon} \sin s\,\arctan x(s)\,ds = O(s) + \pi\cos\varepsilon,$$

hence

$$f(b) = \pi\,(1 - 2\cos C) + O(\varepsilon).$$

By the shooting method, we are done. □